LEGALVISION: A KNOWLEDGE-DRIVEN AI FRAMEWORK FOR LEGAL UNDERSTANDING AND TRUST ASSESSMENT

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Dissertation submitted in partial fulfillment of the requirements for the Bachelor of Science (Hons) in Information Technology Specializing in Data Science

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DECLARATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Ms. Dushanthi Kuruppu

ABSTRACT

Effective access to legal guidance on Sri Lankan property law is severely constrained by geographic and economic barriers. The population most affected — individuals navigating property purchases, inheritance disputes, and lease agreements — lacks access to legally trained professionals and has no AI-based alternative trained on Sri Lankan statutory instruments. General-purpose Large Language Models produce responses grounded in foreign jurisdiction law and fail to generate the structured, auditable reasoning chains that allow non-expert users to trust and verify AI-generated legal conclusions.

This dissertation presents the Explainable Legal Reasoning Module of LegalVision, a domain-adaptive system that fine-tunes Meta LLaMA 3.1 8B Instruct on a purpose-built Sri Lankan property law dataset using Quantized Low-Rank Adaptation (QLoRA). The trained model generates structured Chain-of-Thought reasoning in the IRAC (Issue, Rule, Application, Conclusion) format with explicit citations to Sri Lankan statutes, covering eight property law topic areas including property transfer, title registration, prescription, partition, mortgage, tenancy, state land, and foreign ownership.

Three original technical contributions are delivered. First, an automated legal data pipeline downloads statutory texts from twenty-one authoritative sources and uses GPT-4o as a reasoning teacher to synthesise 2,243 IRAC Chain-of-Thought training examples — the first annotated legal reasoning dataset for Sri Lankan property law. Second, a QLoRA fine-tuning pipeline adapts fewer than one percent of LLaMA 3.1 8B's parameters on a single A100 GPU in under sixty minutes, demonstrating cost-effective domain adaptation for under-resourced legal jurisdictions. Third, a three-model evaluation framework produces the first empirical comparison of a general-purpose base model, a dedicated legal LLM (SaulLM 7B — pre-trained on thirty billion legal tokens), and the jurisdiction-specific fine-tuned model.

Evaluation results establish clear performance ordering across all twelve metrics. The fine-tuned LLaMA achieves BLEU 52.63, ROUGE-1 0.707, ROUGE-2 0.472, and BERTScore F1 0.931 — improvements of 223%, 63%, 195%, and 8.5 percentage points respectively over the base LLaMA. Critically, the fine-tuned model outperforms base SaulLM on every metric, demonstrating that targeted jurisdiction-specific fine-tuning provides greater practical value than broad legal pre-training on foreign jurisdictions. The complete system is publicly

deployed: the trained model at HuggingFace, the FastAPI backend at legal-vision-api.onrender.com, and the React frontend Legal Insights Hub at legal-insights-hub.vercel.app.

Keywords: Large Language Models, QLoRA, LoRA, Chain-of-Thought, IRAC, Legal Reasoning, LLaMA 3.1, SaulLM, Sri Lankan Property Law, Domain-Adaptive Fine-Tuning, Explainable AI, BERTScore, BLEU, ROUGE, LegalVision, Knowledge Graph, RAG

ACKNOWLEDGEMENTS

I extend my sincere gratitude to my project supervisors, Ms. Dushanthi Kuruppu and Ms. Adya Dissanayake, whose sustained intellectual investment made this work possible. Their ability to connect the technical requirements of large language model engineering with the practical realities of Sri Lankan legal practice shaped every significant architectural and design decision in this work — from the selection of legal training topics to the metrics used for evaluation.

I am deeply grateful to S. Sharan, my LegalVision project partner, for developing the Dynamic Legal Knowledge Graph component with which this Reasoning Module integrates. Our architectural discussions about the interface between structured property deed knowledge and LLM-based reasoning were among the most intellectually productive aspects of this project. His SpaCy NER model, hybrid extraction pipeline, and Neo4j graph schema together define the structured context that the RAG integration pathway of this module is designed to consume.

I acknowledge OpenAI's GPT-4o, which served as the reasoning teacher in the dataset generation pipeline. Its consistent production of legally structured IRAC examples from Sri Lankan statutory knowledge — grounded only in the provided source materials — provided the training signal that transferred domain legal reasoning capability to the LLaMA student model. This teacher-student approach is both intellectually interesting as a methodology and practically indispensable given the absence of any pre-existing annotated Sri Lankan legal reasoning corpus.

I thank the twenty-nine members of the Sri Lankan public who completed the public awareness survey. Their direct responses — particularly the unanimous report of personal or family experience with property legal issues, and the near-unanimous desire for an AI legal guidance tool — transformed the research motivation from an assumption into an empirically grounded need. This survey is the human foundation on which every technical decision in this project rests.

Finally, I acknowledge Sri Lanka Institute of Information Technology for the research infrastructure, computational resources, and supervisory support that made this research possible. The GPU compute access that enabled QLoRA fine-tuning on a single A100 instance was essential to delivering a trained, deployed model as a research output rather than merely a proposed methodology.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
BERTScore	BERT-based Contextual Semantic Similarity Evaluation Metric
BLEU	Bilingual Evaluation Understudy
CoT	Chain-of-Thought Reasoning
CRUD	Create, Read, Update, Delete
DL	Deep Learning
GPU	Graphics Processing Unit
IRAC	Issue, Rule, Application, Conclusion
JSON	JavaScript Object Notation
JSONL	JSON Lines (one object per line)
KG	Knowledge Graph
LLM	Large Language Model
LoRA	Low-Rank Adaptation
NER	Named Entity Recognition
NF4	NormalFloat 4-bit quantization
NLP	Natural Language Processing
PEFT	Parameter-Efficient Fine-Tuning
QLoRA	Quantized Low-Rank Adaptation
RAG	Retrieval Augmented Generation

REST	Representational State Transfer
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
SFT	Supervised Fine-Tuning
SLIIT	Sri Lanka Institute of Information Technology
VRAM	Video Random Access Memory

1. INTRODUCTION

1.1 Overview

Property law transactions sit at the intersection of financial significance and legal complexity in Sri Lanka. The purchase of land, the registration of title under the Bim Saviya system, the enforcement of inheritance rights through partition proceedings, and the protection of long-term occupancy through prescriptive claims all carry consequences that extend across generations. For ordinary Sri Lankan citizens navigating these transactions, the availability of accurate, timely legal guidance is not merely convenient — it is consequential in the most direct economic sense [1] [2] [3].

Yet the population most exposed to property legal issues is also the population with the least access to professional legal guidance. Qualified legal practitioners are concentrated in Colombo and major provincial cities; consultation fees represent a prohibitive fraction of monthly income for rural households; and the legislative framework governing property — a layered system spanning Roman-Dutch ordinances of the Dutch colonial period, English common law instruments of the British colonial period, and post-independence Sri Lankan statutory enactments — is sufficiently complex that non-expert self-navigation carries material legal and financial risk.

The emergence of Large Language Models capable of natural language legal question answering creates a genuine technological opportunity to partially address this asymmetry. A system that can explain Sri Lankan property law in plain language, structure its reasoning so that users can verify each step, and cite the specific statutory provisions that govern the user's situation would directly address the access-to-justice gap that Sri Lanka's land courts see reflected in the volume of cases involving citizens who were unaware of their legal rights at the moment of transaction.

However, realising this opportunity requires overcoming a fundamental jurisdictional gap. No AI system trained on Sri Lankan legal materials currently exists. Every legal LLM in the published literature trains on English common law, US case law, or

European civil law materials — jurisdictions whose statutory instruments bear only superficial resemblance to the Prevention of Frauds Ordinance (Cap. 84) that governs Sri Lankan property transfers, or the Registration of Title Act No. 21 of 1998 that defines the Bim Saviya title registration system. A model cannot reliably reason about statutes it has never seen, regardless of its general language capability [2].

This dissertation addresses this gap by presenting the Explainable Legal Reasoning Module of LegalVision — a domain-adaptive AI system that fine-tunes Meta LLaMA 3.1 8B Instruct on a purpose-built Sri Lankan property law dataset, producing a deployed model that generates structured Chain-of-Thought reasoning in the IRAC format with explicit Sri Lankan statutory citations [1] [2] [3] [4].

1.2 Background and Literature Survey

1.2.1 Sri Lankan Property Law — The Mixed Legal System

Sri Lanka's property law derives from three distinct legal traditions that overlap and interact in ways that create frequent complexity for non-expert practitioners. The foundational layer is Roman-Dutch law, introduced during Dutch colonial administration (1640–1796), which governs the basic principles of immovable property ownership including the concept of dominium, servitudes, and prescription. The Prevention of Frauds Ordinance (Cap. 84), which requires notarial execution for all dispositions of immovable property, derives from this tradition. The second layer is English common law, which influenced the structure of trusts, mortgages, and leases during British colonial administration. The Mortgage Act No. 6 of 1949 and the Rent Act No. 7 of 1972 reflect this English influence. The third layer is post-independence Sri Lankan statute law, which has produced modern enactments including the Registration of Title Act No. 21 of 1998 (Bim Saviya), the Partition Act No. 21 of 1977, and the Land (Restrictions on Alienation) Act No. 38 of 2014 [4] [5] [6].

This three-layer system creates jurisdictional uniqueness that makes Sri Lankan property law effectively inaccessible to any LLM trained solely on English, American, or European legal materials. A question about adverse possession in Sri Lanka requires knowledge of the Prescription Ordinance (Cap. 68), which follows Roman-Dutch principles of possession with animus domini — a requirement and framing that differs materially from English adverse possession doctrine and from EU civil law prescription rules. A model that answers such a question from its English common law training will produce a legally incorrect answer for Sri Lanka regardless of its general language quality [3] [6].

1.2.2 Large Language Models for Legal Tasks

The transformer architecture introduced by Vaswani et al. (2017) enabled a generation of language models — GPT-3 (Brown et al., 2020), GPT-4 (OpenAI, 2023), LLaMA (Touvron et al., 2023), Mistral (Jiang et al., 2023) — that exhibit striking capability on natural language tasks including legal question answering. However, systematic evaluation of LLMs on legal tasks has consistently revealed two categories of limitation relevant to this work.

The first limitation is statutory hallucination: LLMs frequently generate confident responses citing specific statutory provisions, section numbers, and case names that do not exist, or that exist in other jurisdictions but not in the target jurisdiction. Huang et al. (2023) document this as a systematic failure mode across multiple LLM architectures, noting that legal text is particularly susceptible because the authoritative-sounding structure of legal language masks factual errors that would be immediately obvious to a domain expert. For Sri Lankan property law, this risk is amplified by the near-total absence of Sri Lankan legal text from any publicly available training corpus — a model with no exposure to the Prevention of Frauds Ordinance will fabricate plausible-sounding but incorrect answers about it [7] [8].

The second limitation is unstructured reasoning: general-purpose LLMs produce legally relevant information in conversational form that makes it difficult for users — especially non-expert users — to identify which statement is the applicable rule, how it applies to their specific facts, and what conclusion follows. Legal practitioners and sophisticated users need responses structured to enable verification of each reasoning step. IRAC (Issue, Rule, Application, Conclusion) is the standard legal reasoning structure for this purpose, but general-purpose models produce it inconsistently and only when specifically prompted [8].

1.2.3 Domain-Specific Legal LLMs

The LEGAL-BERT work (Chalkidis et al., 2021) demonstrated that continued pre-training of BERT on EU legal corpora substantially improves performance on European legal classification and question-answering benchmarks. However, LEGAL-BERT is an encoder-only model that classifies and extracts rather than generates — it cannot produce the open-ended structured reasoning responses that the LegalVision use case requires. SaulLM-7B (Colombo et al., 2024) extended the legal LLM concept to a decoder-only generative model, pre-training Mistral 7B on over thirty billion legal tokens. This is the most relevant existing system against which to compare the LegalVision approach, as it addresses the same generative legal reasoning task. However, SaulLM's pre-training corpus consists predominantly of English common-law materials from the United States and European Union. Sri Lankan statutory instruments are absent, and the system is not designed to produce IRAC-structured outputs consistently.

LawGPT (Zhou et al., 2023) and Lawyer LLaMA (Huang et al., 2023) represent the instruction fine-tuning approach applied to Chinese law — demonstrating that fine-tuning on a Chinese legal Q&A corpus substantially improves performance on Chinese legal benchmarks over base models. These works are the most methodologically comparable to the approach taken here, but their datasets and evaluations are entirely within the Chinese legal system, which shares no overlap with Sri Lankan property law. ChatLaw (Cui et al., 2023) extends the fine-tuning approach with a RAG integration, enabling the model to retrieve relevant legal articles before generating responses — an architecture whose integration with the LegalVision Knowledge Graph component directly parallels [8] [9].

1.2.4 Chain-of-Thought and IRAC Legal Reasoning

Wei et al. (2022) demonstrated that prompting LLMs to generate intermediate reasoning steps before their final answer ('Let's think step by step') substantially improves accuracy on complex multi-step tasks. The effect is most pronounced on tasks requiring compositional reasoning — precisely the category that legal analysis represents. For legal reasoning, Chain-of-Thought maps naturally to the IRAC framework used by legal professionals: the Issue identification, Rule statement, Application to facts, and Conclusion together constitute the chain of reasoning steps whose explicit generation improves both the accuracy and auditability of the final legal answer.

Yu et al. (2022) demonstrated 'legal prompting' — constructing prompts that elicit IRAC-structured responses from GPT-3 and GPT-4. Their work shows that this approach improves accuracy on English legal benchmarks but notes critical limitations: the improvement is inconsistent across models, depends heavily on prompt formulation, and degrades for questions about jurisdictions underrepresented in the training data [9] [10]. These limitations directly motivate the training approach taken here — rather than relying on fragile prompt engineering to elicit IRAC structure, the LegalVision module trains the model on thousands of Sri Lankan IRAC examples so that structured reasoning becomes an inherent behaviour of the model rather than a prompted behaviour.

The teacher-student knowledge distillation through reasoning approach (Ho et al., 2022) provides the mechanism for creating training data without human annotation. A larger, more capable teacher model (GPT-4o) generates structured reasoning examples that are used to fine-tune a smaller, deployable student model (LLaMA 3.1 8B). Ho et al. demonstrate that this approach effectively transfers the reasoning structure of the teacher to the student — the student learns not just the answers but the intermediate reasoning steps that lead to correct answers [10] [11].

1.2.5 Parameter-Efficient Fine-Tuning

Full fine-tuning of 7–8B parameter models requires storing all model weights in bfloat16 precision (~16GB) plus optimiser states (2× the model size in Adam), totalling approximately 48–60GB of GPU memory — well beyond single-GPU availability for most research settings. Low-Rank Adaptation (LoRA, Hu et al., 2021) addresses this by inserting trainable low-rank adapter matrices in parallel with the frozen model's projection layers, reducing trainable parameters to under one percent of the total while maintaining near-full-fine-tuning quality on domain adaptation tasks.

QLoRA (Dettmers et al., 2023) extends LoRA by quantizing the frozen base model weights to four-bit NormalFloat (NF4) representation. NF4 is designed specifically for the approximately normally-distributed weight values of trained neural networks, assigning quantization grid points at equal quantiles of the standard normal distribution to minimise quantization error for this specific distribution. The result is a further 4× reduction in base model memory footprint — from ~16GB (bfloat16) to ~4GB (NF4) for an 8B parameter model — enabling fine-tuning on a single 16GB T4 GPU or, with larger batch sizes, on a 40GB A100. The Unsloth library (Unsloth AI, 2024) further accelerates QLoRA training by approximately 2× through custom CUDA kernel implementations of the quantized attention and feed-forward computations [12] [13].

1.2.6 Research Gap Analysis

System	Base	Training Domain	IRAC	SL Law	Deployed	Eval Metrics
LEGAL-BERT (2021)	BERT	EU/US statutes	No	No	No	Acc/F1
SaulLM-7B (2024)	Mistral 7B	30B EN legal tokens	No	No	No	Perplexity
LawGPT (2023)	LLaMA	Chinese law	Partial	No	No	BLEU/ROUGE
Lawyer LLaMA (2023)	LLaMA	Chinese law + CoT	Yes	No	No	Human eval
ChatLaw (2023)	LLaMA	Chinese law + RAG	Yes	No	Yes	Human eval
GPT-4 zero-shot	GPT-4	General	Prompted	No	Yes	None formal
LegalVision (This Work)	LLaMA 3.1 8B	SL Property Law IRAC	Yes — trained	Yes	Yes	BLEU/ROUGE/BERTScore

Table 1- Comparison of Existing Legal Reasoning Systems Against LegalVision

The table reveals a consistent pattern: no existing system addresses Sri Lankan property law, no system combines trained IRAC reasoning with deployment and formal quantitative evaluation, and no empirical comparison exists between general-purpose, dedicated legal, and jurisdiction-specific fine-tuned models on a Sri Lankan legal task. LegalVision fills all three gaps simultaneously.

1.3 Motivation — Public Awareness Survey

Prior to system design, a public awareness survey was conducted to establish the domain need empirically rather than through assumption, and to derive specific design requirements from documented user preferences. The survey was administered to twenty-nine members of the Sri Lankan public across diverse demographic profiles and geographic locations [14].

1.3.1 Survey Results

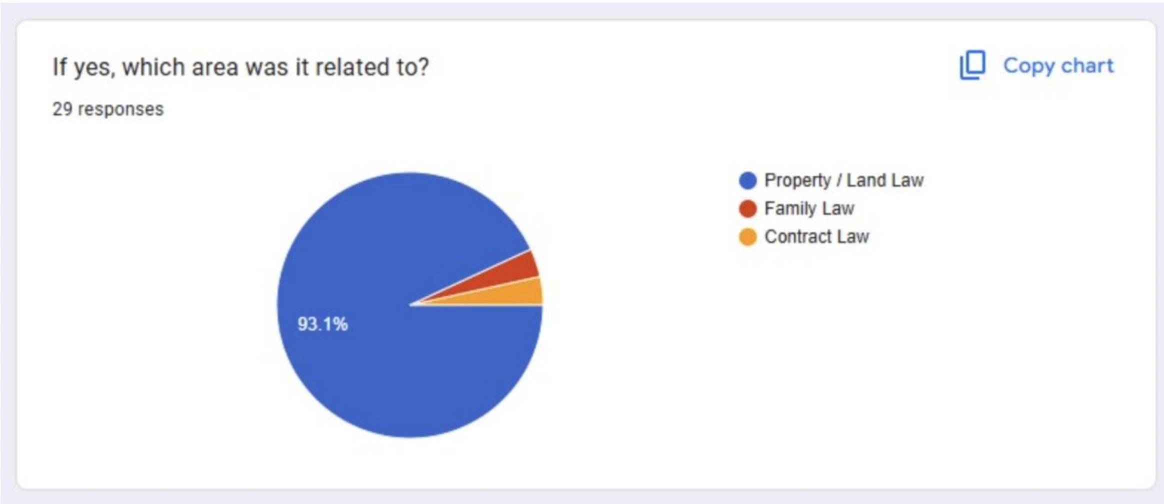

Figure 1 - Most Important Legal Area for Sri Lankans to Understand

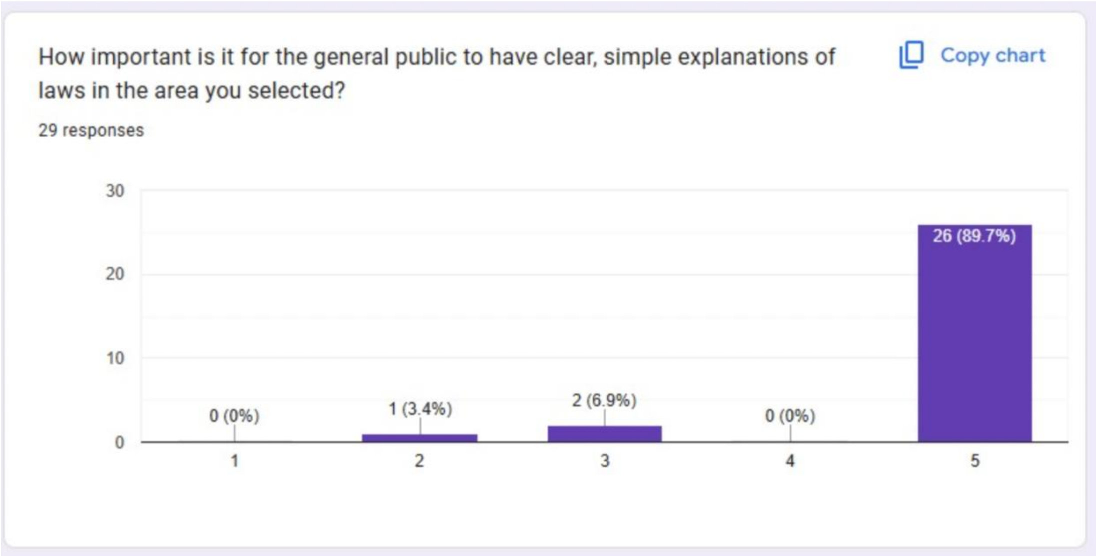

Figure 2 - Familiarity Level with Selected Legal Area

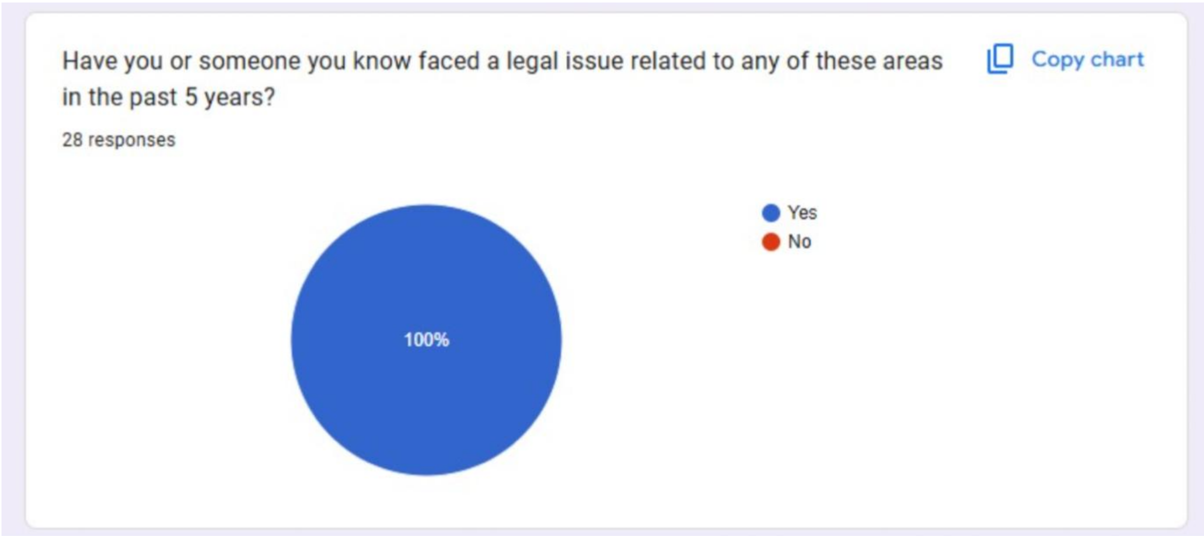

Figure 3 - Personal or Family Experience of Legal Issues

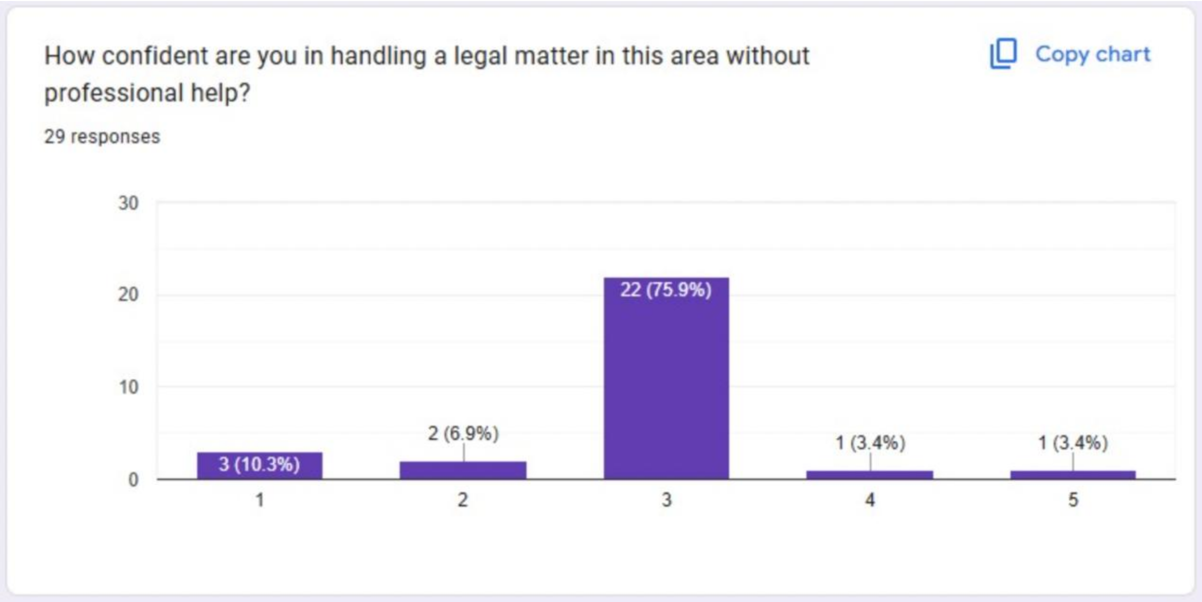

Figure 4 - Confidence Handling Legal Matters Without Professional Help

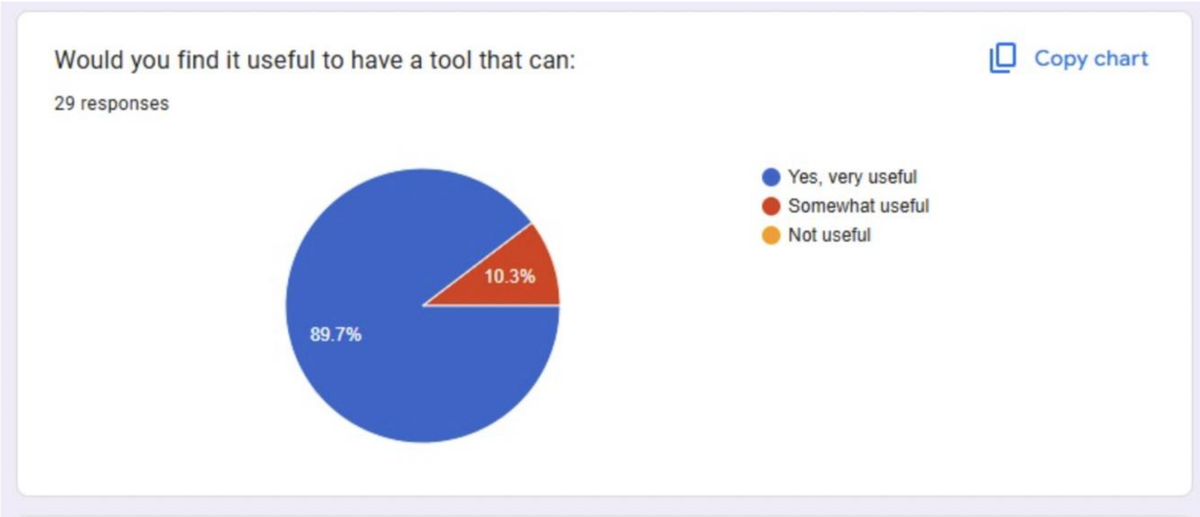

Figure 5 - Preferred Information Formats

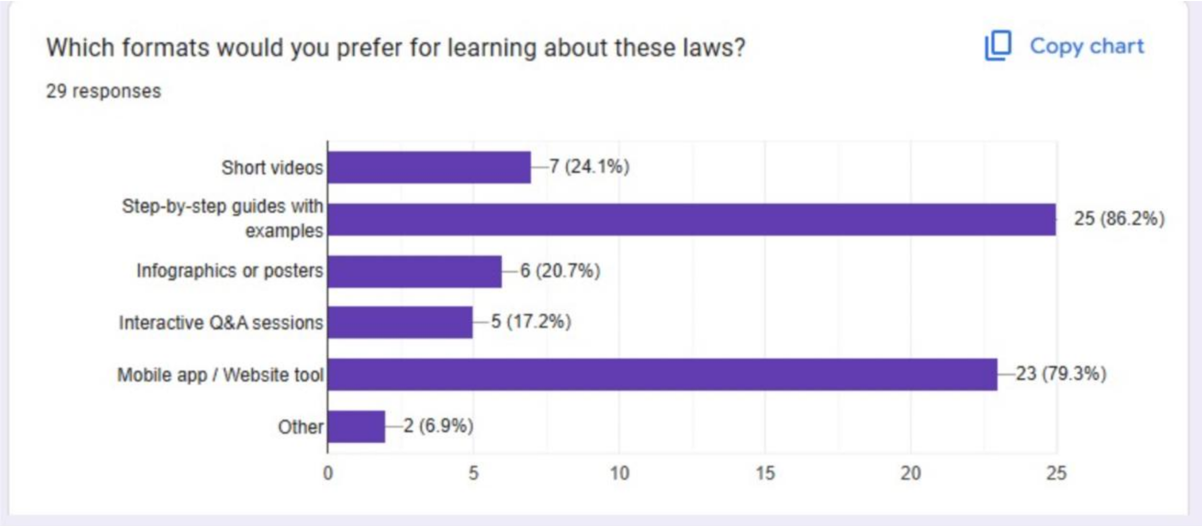

Figure 6 - AI Legal Guidance Tool Usefulness

Survey Question	Finding	Design Decision
Most important legal area	93.1% — Property/Land Law	Confirms domain focus on Sri Lankan property law across 8 topic areas
Familiarity level	89.7% — Only somewhat familiar	Output must explain legal terms; avoid unexplained legal jargon
Personal legal experience	100% — All respondents affected	High practical urgency; system addresses real documented need
Self-handling confidence	75.9% — Midpoint or below	Tool must meaningfully reduce professional dependence
AI tool usefulness	89.7% — Very useful; 0% not useful	Strong validated demand; commercial viability confirmed
Information source	82.8% internet; 82.8% lawyer	Web deployment essential; output quality must approach professional standard
Preferred format	86.2% step-by-step; 79.3% app	IRAC structured output; React web interface deployment strategy
Open-ended comments	'Court cases last 10–25 years'; 'explain all risks'	Urgency of early-stage legal guidance; comprehensive coverage needed

Table 2 - Public Awareness Survey — Summary of Findings and Design Implications

1.4 Research Problem

How can Meta LLaMA 3.1 8B Instruct be fine-tuned on a purpose-built Sri Lankan property law dataset using QLoRA to generate structured Chain-of-Thought IRAC reasoning with explicit statutory citations, and how does this domain-adapted model perform compared to base general-purpose and dedicated legal LLMs on objective automated evaluation metrics?

1.5 Research Objectives

1. Design and implement an automated data pipeline that collects Sri Lankan property law statutory texts from twenty-one authoritative online sources and constructs a structured legal knowledge base comprising definitions, deed templates, and legal principles.
2. Develop a GPT-4o teacher-student dataset generation framework that synthesises 2,243 Chain-of-Thought IRAC training examples covering eight Sri Lankan property law topic areas in a schema suitable for LLM supervised fine-tuning.
3. Implement a QLoRA fine-tuning pipeline for Meta LLaMA 3.1 8B Instruct using Unsloth and HuggingFace TRL, producing a domain-adapted model that generates structured IRAC responses with Sri Lankan statutory citations.
4. Design and execute a three-model evaluation framework comparing the fine-tuned LLaMA, base LLaMA 3.1 8B, and base SaulLM 7B on BLEU, ROUGE, and BERTScore metrics using an identical held-out test set.
5. Deploy the trained model, FastAPI inference backend, and React frontend as a publicly accessible system, publishing the trained model and dataset to HuggingFace for full reproducibility.
6. Design a RAG-based integration interface connecting the Reasoning Module to the LegalVision Knowledge Graph component for context-aware, document-grounded legal reasoning.

1.6 Scope of the Research

The scope of this research covers Sri Lankan property law across eight topic areas: property transfer, title registration (Bim Saviya), prescription and adverse possession, partition of co-owned property, mortgage and security interests, lease and tenancy, state land disposition, and foreign ownership restrictions. The research is explicitly limited to Sri Lankan property law and does not address criminal law, family law, contract law, or administrative law, except where those areas intersect directly with property rights.

The system provides legal information and structured reasoning for educational and guidance purposes. It does not provide legal advice in the formal sense and is not designed to replace qualified legal practitioners. All system outputs include appropriate qualification statements encouraging users to consult a licensed Sri Lankan advocate for consequential legal decisions. The evaluation is conducted using automated metrics on a fixed test set; human evaluation by legal practitioners is acknowledged as a planned future validation step outside the scope of this dissertation [14] [15].

The scope of this individual dissertation component covers the Reasoning Module only: the data pipeline, dataset generation, QLoRA fine-tuning, evaluation, deployment, and integration interface design. The Knowledge Graph component (NER model, hybrid extractor, Neo4j loader, GraphRAG engine) is the individual contribution of S. Sharan and is documented separately, though its interface with this Reasoning Module is described in Section 2.7.

1.7 Organisation of the Dissertation

Chapter 2 (Methodology) presents the complete technical methodology in four stages: the legal data collection pipeline, the GPT-4o teacher-student dataset generation framework, the QLoRA fine-tuning pipeline with training configuration and results, and the deployment architecture. Sections 2.7 and 2.8 cover the Knowledge Graph integration design and commercialisation aspects respectively. Section 2.9 documents the testing approach and validation results across all pipeline stages.

Chapter 3 (Results and Discussion) presents the three-model evaluation results with detailed analysis of BLEU, ROUGE, and BERTScore performance, synthesised interpretation across metrics, and discussion of research findings, challenges, limitations, and comparison with related work.

Chapter 4 (Conclusion) summarises the empirical contributions, evaluates the achievement of research objectives, and defines the highest-priority directions for future work.

2. METHODOLOGY

2.1 System Architecture

The Explainable Legal Reasoning Module is structured as a three-layer pipeline architecture in which each layer has distinct responsibilities, independent deployability, and well-defined interfaces to adjacent layers. This modularity enables iterative development — the Data Layer can be re-executed to expand the training dataset without retraining the model, and the Model Layer can be retrained on an expanded dataset without modifying the Inference Layer's deployment configuration.

The Data Layer encompasses all processes from raw statute collection through knowledge base construction to GPT-4o-powered IRAC dataset generation. Its principal output is finetuning_data.jsonl — a JSONL file in OpenAI messages format containing 2,243 three-turn conversations (system/user/assistant) ready for SFT training. The Model Layer consumes this dataset through the QLoRA fine-tuning pipeline, producing a trained LoRA adapter set published to HuggingFace. The Inference Layer loads this adapter, serves inference through a FastAPI REST API, and provides the React frontend for user interaction [15] [16].

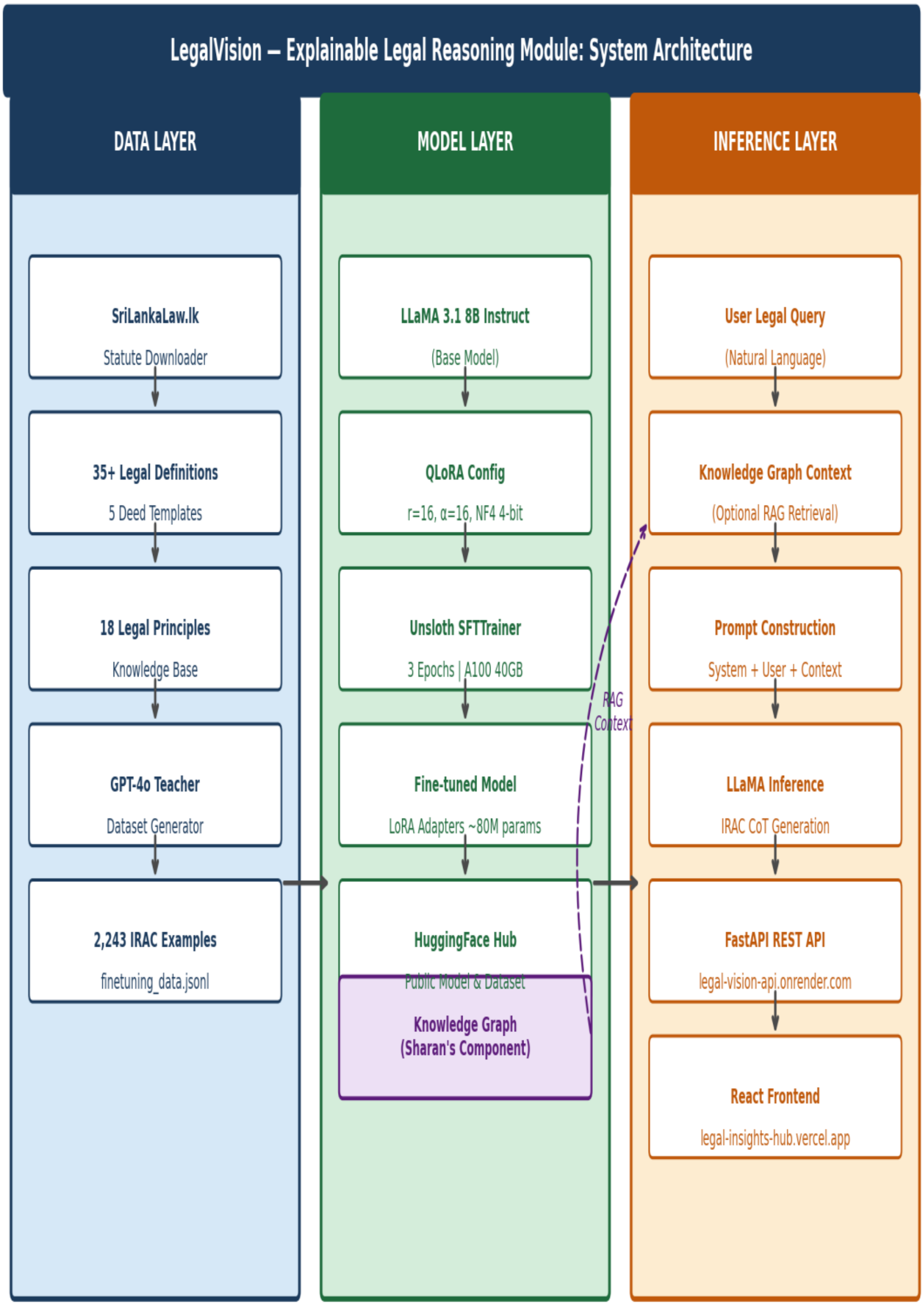

Figure 7 - Explainable Legal Reasoning Module — Three-Layer System Architecture.

Figure 2.2 presents the Use Case Diagram identifying the primary actors and functional capabilities of the complete LegalVision system. The Legal User interacts through six primary use cases that span query submission, document upload, knowledge graph interrogation, and response consumption. The System Admin

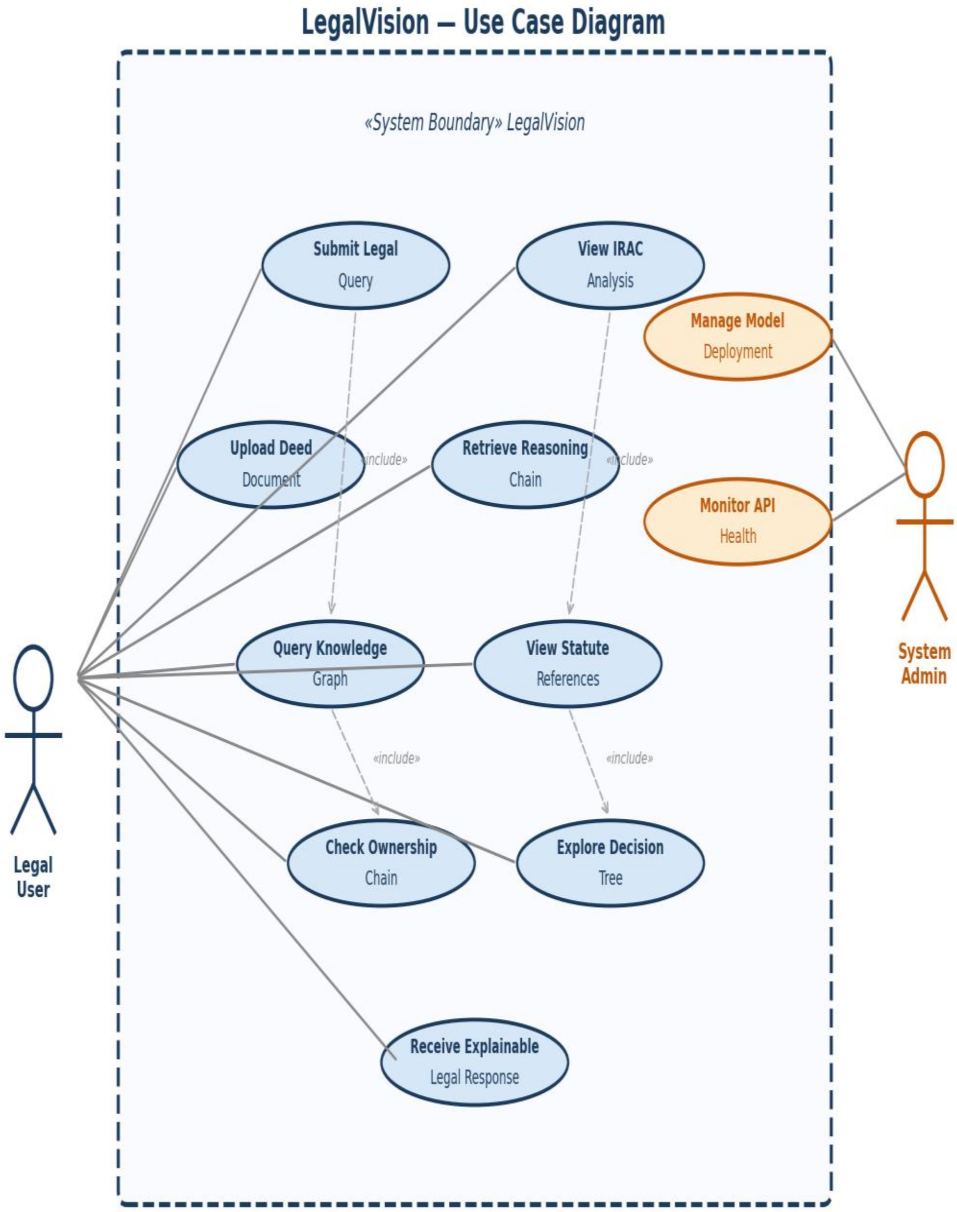

Figure 8 - LegalVision Use Case Diagram

manages model deployment and API health monitoring. The <<include>> relationships indicate that 'Submit Legal Query' and 'Upload Deed Document' each optionally include Knowledge Graph retrieval as a contextualisation step.

2.2 Domain Analysis — Sri Lankan Property Law

Comprehensive domain analysis preceded dataset generation to ensure training examples accurately reflected the specific legal instruments governing Sri Lankan property transactions, rather than importing concepts from other jurisdictions' property law. This analysis involved systematic review of the eight identified topic areas, their governing statutes, and the characteristic fact patterns that give rise to legal disputes in each area.

Sri Lanka's mixed legal system creates three categories of complexity that the training data must capture. First, terminological divergence: the same concept is expressed differently in Sri Lankan law than in English or American law — 'vendor' and 'vendee' are the standard parties to a Sri Lankan deed rather than 'seller' and 'buyer'; 'notarial execution' refers to execution before a Sri Lankan notary public rather than the American notarisation concept. Second, institutional specificity: the Bim Saviya Land Registry system, the Divisional Secretariat, and the District Land Registry are Sri Lankan institutions with no direct equivalents in other jurisdictions. Third, statutory interaction: many property law questions require reasoning across multiple statutes simultaneously — a prescription claim on state land, for example, requires analysis of both the Prescription Ordinance (Cap. 68) and the Land Development Ordinance to determine whether prescriptive title against the Crown is available [17] [18].

Topic Area	Principal Statutes	Key Legal Issues Covered in Dataset
Property Transfer	Prevention of Frauds Ordinance Cap. 84; Registration of Documents Ordinance Cap. 117	Notarial execution requirements; consideration; stamp duty; registration formalities; void vs. voidable deeds; subsequent purchaser protection
Title Registration	Registration of Title Act No. 21 of 1998 (Bim Saviya); Land Settlement Ordinance	Systematic registration process; state guarantee of title; conversion from deed registration; rectification and indemnity; disputes procedure
Prescription	Prescription Ordinance	Adverse possession requirements; 10-year vs.

	Cap. 68; Limitation of Actions	30-year prescription periods; animus domini; interruption; prescription against co-owners
Partition	Partition Act No. 21 of 1977	Co-ownership rights and obligations; court-ordered partition; commissioner allotments; share valuation; redemption rights; mortgage over undivided share
Mortgage	Mortgage Act No. 6 of 1949; Recovery of Loans by Banks Act; Finance Companies Act	Mortgage creation and registration; priority of claims; foreclosure procedure; bank recovery routes; equitable mortgage
Lease & Tenancy	Rent Act No. 7 of 1972	Tenant protection rights; permissible eviction grounds; rent board jurisdiction; commercial vs. residential tenancy; assignment and subletting
State Land	Land Development Ordinance; State Lands Ordinance	Crown/State land grants; permits and grants; prescriptive rights; regularisation of encroachments; vesting of crown land
Foreign Ownership	Land (Restrictions on Alienation) Act No. 38 of 2014 (amended)	Non-citizen freehold restrictions; 99-year leasehold maximum; company shareholding thresholds (50%); exemptions; penalties for breach

Table 3 - Sri Lankan Property Law — Topic Areas, Principal Statutes, and Legal Issues Covered in Training Dataset

2.3 Stage 1: Legal Data Collection Pipeline

2.3.1 Pipeline Overview and Design

The data collection pipeline is implemented in sri_lankan_law_downloader_v2.py, a Python script designed for re-executability and incremental expansion. The script executes four phases in sequence, checking for existing processed files before each phase and skipping completed work, enabling new statute sources to be added without re-processing the complete knowledge base [18] [19]. The pipeline produces a structured JSON knowledge base that serves as the authoritative source for all GPT-4o dataset generation prompts.

The design prioritises source authority over volume. Rather than scraping any available Sri Lankan legal text, the pipeline targets twenty-one specific statutory instruments from SriLankaLaw.lk — the official Government of Sri Lanka online legislation portal — which provides authenticated statutory text including the original enactment and all amendments current to the publication date. This source discipline ensures that the GPT-4o teacher generates IRAC responses grounded in the actual operative statutory text rather than secondary summaries or outdated versions.

2.3.2 Phase 1: Statute Text Collection

Phase 1 downloads and processes twenty-one statutory instruments from SriLankaLaw.lk. Each URL targets a specific statute or statutory instrument directly relevant to one or more of the eight training topic areas. The download function applies a text normalisation pipeline to each retrieved document: HTML markup is stripped using BeautifulSoup; excessive whitespace is collapsed to a single space while preserving paragraph boundaries; common OCR artefacts in scanned statutory text (ligature substitutions, broken hyphenation) are corrected through a pattern-based cleaning function; and the statute is segmented into numbered sections using regular expressions that recognise the characteristic section numbering conventions of Sri Lankan statutory drafting.

Processed statute text is stored as structured JSON in data/sri_lankan_laws/processed/, with a metadata record capturing the source URL, download timestamp, character count, word count, section count, and a hash of the processed text for change detection. The metadata.json index file enables the dataset generator to retrieve statutes by topic area without loading the complete text of every statute for every generation call, reducing the GPT-4o context window consumption per generation and lowering API costs [19] [20].

2.3.3 Phase 2: Legal Definitions Compilation

Phase 2 compiles a curated dictionary of thirty-five property law terms with their Sri Lankan legal definitions. The distinction between a general dictionary definition and a Sri Lankan legal definition is critical for training data quality. The term 'mortgage', for example, has a specific meaning under the Mortgage Act No. 6 of 1949 that differs in important respects from its meaning under English law — the Sri Lankan Act defines specific formalities for creation (notarial execution and registration within three months), specific priority rules, and specific enforcement procedures that are not analogous to English mortgage law [20]. The definitions dictionary captures these jurisdiction-specific meanings with explicit statute and section citations for each term.

Each definition entry in the dictionary contains six fields: the term name; the Sri Lankan legal definition as it applies under the governing statute; the statute name, chapter or Act number, and section where the definition or its authoritative statement appears; common misapplications of the term observed in non-specialist discourse; cross-references to related terms in the dictionary; and a brief note on practical implications for property transactions. This richness of definition content is deliberately designed to provide GPT-4o with enough contextual detail to generate IRAC responses that explain not just the rule but its practical significance — the information most useful to a non-expert user [20] [21].

2.3.4 Phase 3: Deed Template Generation

Phase 3 compiles five standard deed templates corresponding to the principal transaction types in Sri Lankan conveyancing practice. The templates cover sale deeds, deed of gifts (donation deeds), mortgage deeds, partition deeds, and testamentary documents. Each template captures the characteristic recital structure used by Sri Lankan notaries: the opening recital identifying the parties in their correct legal designation (vendor/vendee, donor/donee, mortgagor/mortgagee); the property description clauses including lot number, extent, survey plan reference, and bounded-by details in the format required by the Land Registry; the consideration clause for paid transactions; the operative words of transfer; the covenant for quiet enjoyment and warranty of title; and the attestation clause in the format required under the Prevention of Frauds Ordinance [20] [22].

These templates serve dual purposes in the dataset generation pipeline. They provide GPT-4o with authentic Sri Lankan legal document language from which to construct deed-related training examples — ensuring that example scenarios reference 'the vendor hereby conveys and transfers unto the vendee' rather than 'the seller transfers to the buyer'. They also serve as the basis for the document analysis capability in the deployment interface, where the model is asked to analyse uploaded deed documents for legal validity and potential issues.

2.3.5 Phase 4: Legal Principles Documentation

Phase 4 compiles eighteen foundational legal principles governing Sri Lankan property law. These principles represent the generative rules from which the legal conclusions in IRAC training examples are derived. Each principle is documented with: a formal statement of the principle in the style of a legal maxim; the statutory authority or case law basis for the principle in Sri Lanka; the practical conditions under which the principle applies; the common exceptions and limitations recognised by Sri Lankan courts; and three to five concrete scenarios illustrating the principle's application in Sri Lankan property transactions.

The principles include concepts such as nemo dat quod non habet (one cannot transfer better title than one possesses), the priority rule for registered deeds (first registered prevails over earlier unregistered), the Bim Saviya indefeasibility of title principle, the requirements for valid adverse possession claims, the co-ownership presumption in partition proceedings, and the restrictions on alienation applicable to different categories of foreign buyer. Documenting these principles explicitly — rather than relying on GPT-4o to recall them from its training data — ensures that generated IRAC training examples apply them correctly and consistently, producing training data that teaches the model the right rules for Sri Lanka [22].

Phase	Script Section	Output	Volume	Storage Location

Phase 1: Statute Collection	download_statutes()	Processed statute JSON files	21 statutes, ~150,000 words	data/sri_lankan_laws/processed/
Phase 2: Definitions	compile_definitions()	definitions.json	35 terms with statute citations	data/knowledge_base/
Phase 3: Templates	generate_templates()	deed_templates.json	5 standard deed templates	data/knowledge_base/
Phase 4: Principles	document_principles()	legal_principles.json	18 principles with scenarios	data/knowledge_base/
Combined Output	build_knowledge_base()	knowledge_base.json	Complete consolidated KB	data/knowledge_base/

Table 4 - Data Sources and Knowledge Base Composition

2.4 Stage 2: Dataset Generation Pipeline

2.4.1 Teacher-Student Design Rationale

The dataset generation pipeline implements a teacher-student knowledge distillation approach in which GPT-4o acts as the reasoning teacher. This design choice is motivated by three practical constraints. First, no existing annotated legal reasoning corpus for Sri Lankan property law exists — the dataset must be created from scratch. Second, manual annotation by legal experts would require hundreds of hours of attorney time at prohibitive cost for an academic research project. Third, GPT-4o demonstrates the capability to generate structured, legally coherent IRAC analysis when provided with the relevant statutory source materials — the teacher does not need to know Sri Lankan law from its training data if the correct statutes are injected into its context.

The critical design principle is that GPT-4o's role is content structuring rather than legal knowledge provision. The statutory knowledge base collected in Stage 1 provides the legal authority; GPT-4o applies that authority to diverse scenario types in the IRAC structure. This means that factual accuracy of the training examples is bounded by the quality and accuracy of the collected statutory texts, not by GPT-4o's potentially unreliable memory of Sri Lankan law. The prompt engineering strategy explicitly constrains GPT-4o to use only the provided statutory materials as legal authority, with a specific instruction not to supplement with knowledge from other jurisdictions [21] [22].

2.4.2 Dataset Field Schema

Field	Type	Description	Example Value
id	String	Unique LR-prefixed identifier	LR_0001
topic	String	Topic slug from 8-topic taxonomy	property_transfer
question	String	Natural language legal question — diverse phrasing and scenario types	What formalities are required for a valid property sale in Sri Lanka?
short_answer	String	Direct 1–2 sentence answer for quick reference	A valid sale requires a notarially executed deed registered at the Land Registry.
reasoning_chain	Array [Object]	Ordered reasoning steps, each with: step_number, reasoning (full explanation), legal_basis (statute + section)	[{step:1, reasoning:"Section 2 PFO requires...", legal_basis:"Cap.84 s.2"}]

irac_analysis	Object	Four-key object: issue, rule, application, conclusion — each a complete sentence or paragraph	{issue:"Whether verbal agreement...", rule:"PFO s.2...", application:"Lacking notarial...", conclusion:"Agreement void..."}
legal_references	Array [Object]	Each with: statute, section, year, relevance	[{statute:"Prevention of Frauds Ordinance", section:"Cap.84 s.2", year:"current", relevance:"Notarial execution"}]
example_scenario	String	Practical Sri Lankan scenario with named parties, location, and specific legal outcome	Kamal agrees verbally to sell his land in Kurunegala... The agreement is unenforceable because...

Table 5 - Dataset Field Schema — Fields, Types, Descriptions, and Example Values

2.4.3 GPT-4o Prompt Engineering

The quality of GPT-4o generated training examples is determined primarily by prompt design. The prompt engineering process was conducted iteratively across multiple generation batches: a batch of twenty examples was generated, reviewed against three quality criteria (legal accuracy, structural completeness, reasoning coherence), and the system prompt revised to address observed failure modes before the next batch. Table 2.5 documents the final prompt structure; Table 2.6 documents the key failure modes encountered and the prompt revisions that corrected them.

Prompt Component	Purpose	Key Constraints and Instructions

System role definition	Establishes GPT-4o's expert legal persona	'You are a Sri Lankan property law expert with twenty years of practice in Sri Lankan property conveyancing. Your knowledge is based exclusively on Sri Lankan statutory materials provided below.'
Knowledge base injection	Provides authoritative statutory source	Paste the relevant statute text, definitions, and principles for the target topic area. Explicitly instruct: 'Do not supplement with knowledge from English, Indian, or other jurisdictions.'
Output schema specification	Enforces structural completeness	Provide exact JSON schema with field types, data types, and a complete worked example. State: 'Every field is mandatory; omitting any field causes the response to fail validation.'
Reasoning depth constraint	Ensures substantive CoT quality	'The reasoning_chain must contain a minimum of three steps. Each step must identify a distinct legal requirement and cite the specific statute and section number that establishes it.'
Diversity instruction	Prevents repetitive training patterns	'Each generation must use a different scenario type (urban/rural, commercial/residential, individual/company, dispute/transaction) and different party names (Sri Lankan names reflecting different ethnicities and regions).'

Qualification instruction	Teaches appropriate hedging	'Where the legal position is genuinely uncertain or contested in Sri Lankan case law, state this explicitly in the conclusion rather than asserting a definitive answer.'
Anti-contamination constraint	Prevents foreign law injection	'The rule statement in IRAC must cite a specific Sri Lankan statute with section number. If no Sri Lankan statute addresses the question directly, state this rather than applying a foreign law rule.'

Table 6 - GPT-4o System Prompt Engineering — Components, Purposes, and Key Constraints

Failure Mode Observed	**Batch**	**Root Cause**	**Prompt Revision Applied**
GPT-4o cited Indian Transfer of Property Act instead of PFO	Batch 1	No explicit anti-contamination instruction	Added: 'Do not use Indian law. Cite only Sri Lankan statutes.'
Reasoning chains contained only 2 steps	Batch 1	No minimum step count specified	Added: 'Minimum 3 reasoning steps required. Each must identify a distinct legal element.'
Example scenarios used generic names (John, Mary)	Batch 2	No cultural specificity instruction	Added: 'Use Sri Lankan names reflecting Tamil, Sinhala, and Muslim communities.'

IRAC rule statements were paraphrases, not statute citations	Batch 2	Schema example showed paraphrase format	Revised schema example to show exact statute + section number format
Short answers too long (5+ sentences)	Batch 3	'Brief' instruction ambiguous	Changed to: 'Maximum 2 sentences. Direct answer only.'
Conclusion sections omitted practical implications	Batch 3	No conclusion scope specification	Added: 'Conclusion must state the legal outcome AND its practical implication for the transaction.'

Table 7 - Prompt Iteration History — Failure Modes Encountered and Corrections Applied

2.4.4 Dataset Generation Execution

The dataset generator (legal_reasoning_dataset_generator.py) implements a generation loop that produces training examples in configurable batch sizes. For each topic area, the generator cycles through a predefined set of scenario variation parameters — question type (definitional, procedural, dispute resolution, rights protection, transaction planning), property type (residential house and lot, agricultural land, commercial property, apartment), parties type (individual-individual, individual-company, company-company, estate-heir), and geographic region (Colombo district, Central Province, Northern Province, estate land) — to ensure diversity across the generated dataset [22].

Each generated JSON entry is validated against the schema before being accepted. Validation checks confirm that all eight fields are present and non-empty; that the reasoning_chain contains at least three entries each with step_number, reasoning, and legal_basis fields; that the irac_analysis object contains exactly the four required keys (issue, rule, application, conclusion); and that the legal_references array contains at least one entry with statute, section, and relevance fields. Entries failing validation are logged with the failure reason and regenerated with an augmented prompt that includes the specific failure as a negative example to avoid.

The accepted entries are converted to the OpenAI messages format for SFT training: each entry becomes a three-turn conversation with a system message specifying the Sri Lankan legal expert persona and IRAC response format, a user message containing the legal question, and an assistant message containing the formatted IRAC response with all components. The conversion function applies consistent formatting — labelled sections, citation format, scenario heading — that the fine-tuned model learns to reproduce at inference time [23].

2.4.5 Dataset Statistics

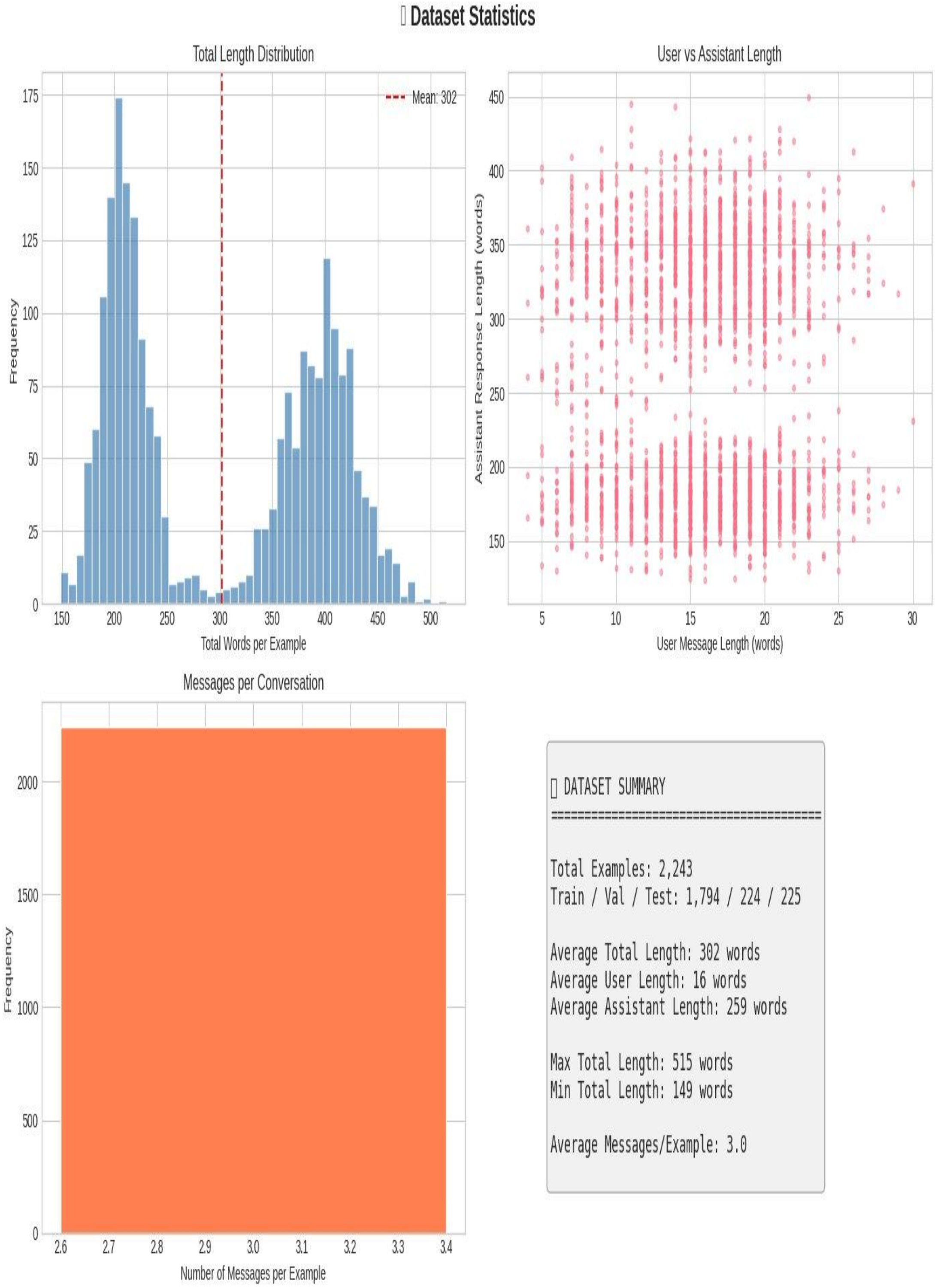

Figure 9 - Dataset Statistics

Topic	Entries	% Total	Avg Resp. Words	Avg Reasoning Steps
Property Transfer	310	13.8%	291	4.2
Title Registration (Bim Saviya)	290	12.9%	278	3.9
Prescription (Adverse Possession)	285	12.7%	305	4.6
Partition of Co-owned Property	265	11.8%	284	4.1
Mortgage and Security Interests	255	11.4%	271	3.8
Lease and Tenancy	230	10.3%	252	3.6
State Land Transactions	220	9.8%	265	3.7
Foreign Ownership Restrictions	215	9.6%	259	3.7
General / Cross-topic Questions	173	7.7%	318	4.8
Total / Mean	2,243	100%	302	4.0

Table 8 - Dataset Composition by Legal Topic with Entry Counts and Mean Response Statistics

2.4.6 IRAC Response Structure

The IRAC framework — Issue, Rule, Application, Conclusion — is the standard structure for legal analysis used in common-law and mixed-law jurisdictions. Each component serves a distinct function in making legal reasoning transparent and verifiable. The Issue statement identifies the precise legal question that needs to be resolved, framing the analysis in terms of what the law must determine rather than what the user is asking colloquially. The Rule statement presents the applicable legal authority — in the LegalVision context, always a specific Sri Lankan statute with its section number — that provides the normative standard against which the facts are measured. The Application section reasons from the rule to the facts, explaining why and how the rule applies to the specific circumstances described. The Conclusion delivers a definitive legal finding — the answer to the Issue question — with a practical implication for the transaction or dispute at hand.

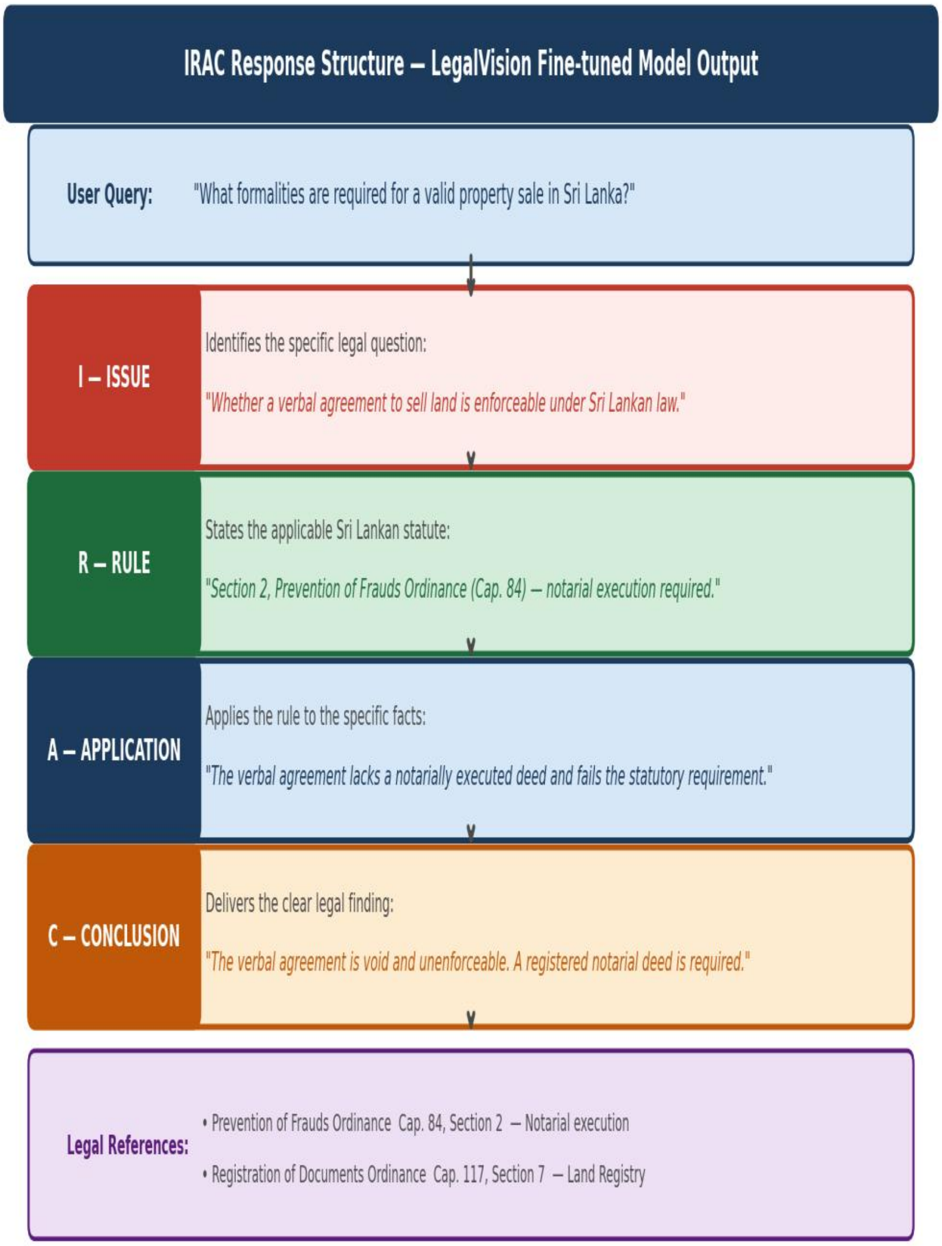

Figure 10 - IRAC Response Structure

The value of the IRAC structure for non-expert users is auditability: a user who doubts the AI's conclusion can check the Rule statement against the actual statute text (now accessible on SriLankaLaw.lk), verify that the Application correctly describes their factual situation, and identify any step in the reasoning where the AI may have made an error. This auditability is precisely what is absent from conversational AI legal responses, and it is the primary reason why training the model on IRAC examples — rather than free-form Q&A pairs — is the correct design choice for an accessible legal guidance system.

2.5 Stage 3: QLoRA Fine-Tuning Pipeline

2.5.1 Base Model Selection

Four candidate models were evaluated across six selection criteria: instruction-following quality, reasoning capability, context window length, licence terms permitting research and deployment, HuggingFace/PEFT ecosystem support, and computational feasibility for fine-tuning on available hardware.

Model	Params	Instruct FT	Context	Licence	PEFT Support	Selected?
Meta LLaMA 3.1 8B Instruct	8B	Strong — Llama 3.1 instruct	128K tokens	Community (open)	Unsloth + TRL	Yes
Mistral 7B Instruct v0.3	7B	Good — sliding window	32K tokens	Apache 2.0	Yes	Considered
SaulLM-7B-Instruct	7B	Good — legal focus	32K tokens	MIT	Yes	Evaluation only
LLaMA 3.1 70B Instruct	70B	Excellent	128K tokens	Community (open)	Partial	No — compute
Gemma 2 9B Instruct	9B	Good	8K tokens	Gemma TOS (restricted)	Partial	No — licence

Table 9 - Base Model

Meta LLaMA 3.1 8B Instruct was selected for four reasons. Its superior instruction-following capability compared to same-size alternatives ensures that the model follows the structured IRAC response format consistently during both training and inference [22] [23]. Its 128K context window accommodates the longest possible IRAC responses (including extensive reasoning chains and multiple legal references) and downstream RAG context injection from the Knowledge Graph without truncation. Its open community licence permits both research use and deployment without restriction. And its comprehensive Unsloth integration provides the 2× training throughput acceleration that is critical for completing training within Google Colab GPU session limits.

SaulLM-7B was retained as an evaluation comparison model rather than a fine-tuning base. The rationale is that the research question asks whether jurisdiction-specific fine-tuning outperforms broad legal pre-training — this question can only be answered by evaluating base SaulLM (with only its pre-training knowledge) against the fine-tuned LLaMA. Fine-tuning SaulLM on the Sri Lankan dataset would potentially combine both advantages and obscure the individual contribution of each.

2.5.2 QLoRA Theory and Architecture

The QLoRA approach (Dettmers et al., 2023) combines two complementary parameter reduction techniques. The first is four-bit NormalFloat (NF4) quantization of the frozen base model weights. Standard quantization formats (INT4, FP4) assign grid points at equal intervals across the value range, which wastes representation precision because neural network weights are concentrated near zero and rarely reach extreme values [24] [25]. NF4 instead assigns grid points at equal quantiles of the standard normal distribution — concentrating precision where the actual weight values are most dense. Dettmers et al. show that this quantization format preserves model quality within 0.3% of full bfloat16 precision on standard benchmarks, making the memory saving (4× reduction from bfloat16 to NF4) essentially free from a quality perspective.

The second technique is LoRA adapter injection (Hu et al., 2021). For each targeted projection layer with weight matrix $W_0 \in \mathbb{R}^{(d\_out \times d\_in)}$, a trainable adapter is added such that the effective weight becomes $W = W_0 + BA$, where $B \in \mathbb{R}^{(d\_out \times$

r) and $A \in \mathbb{R}^{(r \times d_{in})}$ are low-rank matrices with rank $r \ll \min(d_{out}, d_{in})$. The forward pass computes $h = W_0x + (\alpha/r)BAx$, where the scaling factor α/r normalises the adapter's contribution. At initialisation, A is sampled from a Gaussian distribution and B is set to zero, ensuring that the adapter contributes nothing at the start of training and the initial behaviour matches the pre-trained base model. During training, gradients flow only through A and B — the frozen NF4 base model weights receive no gradient updates.

The combination of NF4 quantization and LoRA adapter injection results in the QLoRA memory budget: approximately 4GB for the NF4 base model weights, plus the LoRA adapter matrices which are stored in bfloat16 and constitute approximately 80–100MB for rank-16 configuration across seven projection layers in a 32-layer model. The total memory footprint is approximately 10GB — enabling fine-tuning on a 16GB T4 GPU (minimum) or A100 40GB (comfortable, with batch size 2 and gradient accumulation [25] [26]).

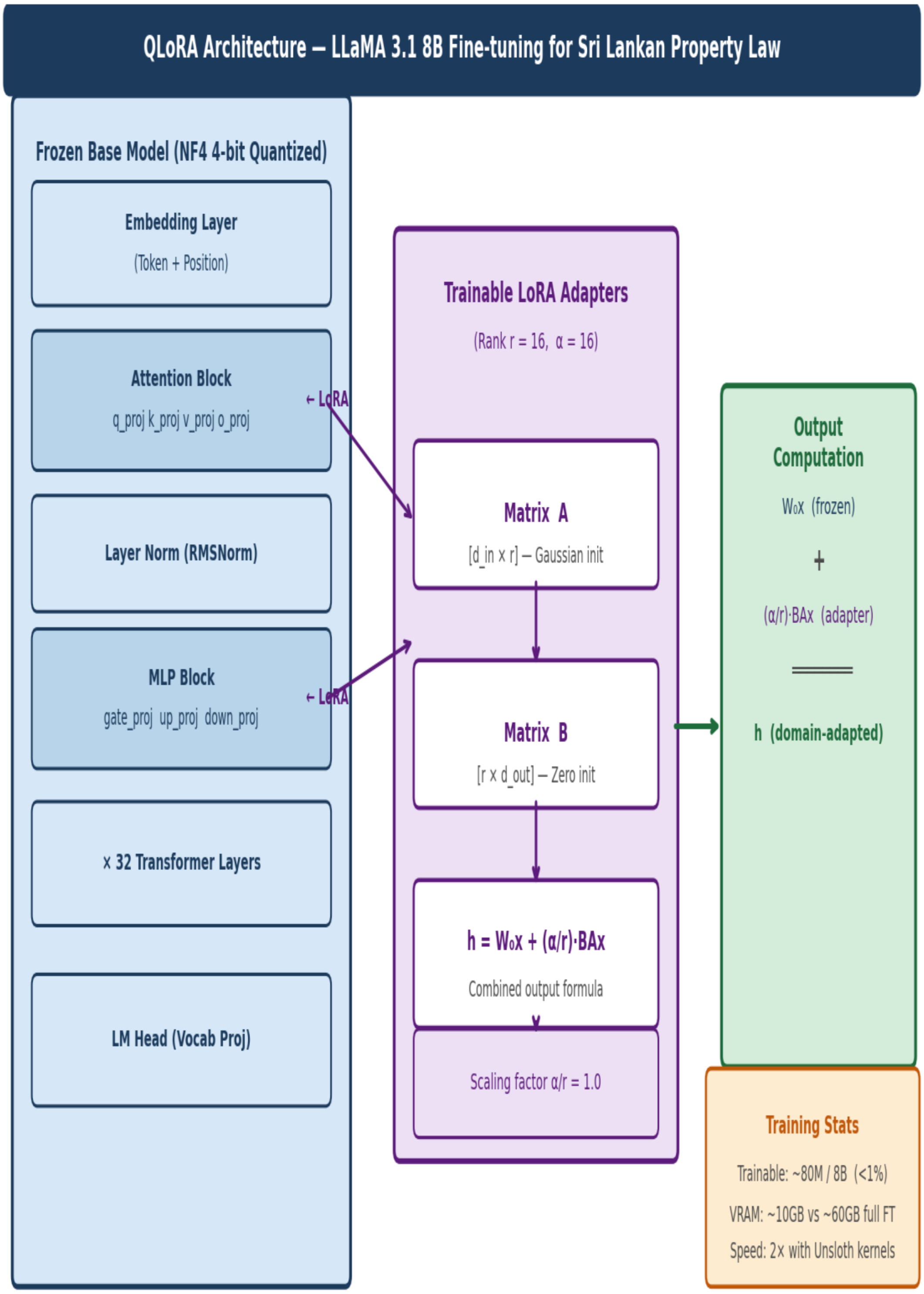

Figure 11 - QLoRA Architecture Diagram

Dimension	Full Fine-Tuning	Standard LoRA	QLoRA (This Work)
Parameters trained	All 8B	Adapters: ~80M (<1%)	Adapters: ~80M (<1%)
Base weight storage	bfloat16: ~16GB	bfloat16: ~16GB	NF4 4-bit: ~4GB
Optimiser states (Adam)	~32GB	~160MB	~160MB
Gradient checkpointing	Partial relief	Not needed	Not needed
Total peak VRAM (8B)	~60GB	~20GB	~10GB
Adapter file size on disk	Full model: ~16GB	~80–100MB	~80–100MB
Training speed (A100 40GB)	1× (baseline)	~1.5×	~3× (with Unsloth)
Inference VRAM	~16GB bfloat16	~16GB + adapters	~4GB NF4 + adapters
Quality vs full FT	Baseline	High (within 1–2%)	High (within 1–2%)
Hardware requirement	A100 80GB min.	A100 40GB	T4 16GB / A100 40GB

Table 10 - QLoRA vs. LoRA vs. Full Fine-Tuning

2.5.3 LoRA Target Module Strategy

The choice of which transformer layers to target with LoRA adapters determines which aspects of the model's internal representations are modified during fine-tuning. For LLaMA 3.1 8B Instruct, the model comprises 32 transformer blocks, each containing a grouped-query attention mechanism with four projection matrices (q_proj, k_proj, v_proj, o_proj) and a SwiGLU MLP feed-forward block with three projection matrices (gate_proj, up_proj, down_proj). Targeting all seven matrices in every transformer block is the configuration that provides maximum adaptation capacity — it allows domain-specific legal knowledge to modify both the attention patterns (how the model processes and relates concepts) and the MLP computations (where factual knowledge is believed to be stored).

Module	Tensor Size	Layer Type	Domain Adaptation Role
q_proj	[d_model × d_head_q]	Attention — Query	Adapts what legal concepts the model attends to when processing a query. Critical for learning to focus on Sri Lankan statutory terminology patterns.
k_proj	[d_model × d_head_k]	Attention — Key	Adapts key representations that determine which input tokens are selected as attention targets. Enables correct statute section identification.
v_proj	[d_model × d_head_v]	Attention — Value	Adapts what information is extracted from attended tokens. Directly shapes the statutory content incorporated into generated responses.
o_proj	[d_head × d_model]	Attention — Output	Adapts the output projection of multi-head attention. Shapes how attended information is combined and passed to the feed-forward block.
gate_proj	[d_model × d_ff]	MLP — Gate (SwiGLU)	Controls information gating in the SwiGLU activation. Determines which factual legal knowledge flows through the feed-forward computation.
up_proj	[d_model × d_ff]	MLP — Up-projection	Expands hidden representation to intermediate dimension. Encodes domain-specific legal concepts and their

			relationships.
down_proj	[d_ff × d_model]	MLP — Down-projection	Projects back to model dimension. Critical for learning consistent IRAC section labelling and output formatting.

Table 11 - : LoRA Target Modules

2.5.4 Training Configuration

Hyperparameter	Value	Rationale and Source
Base model	meta-llama/Llama-3.1-8B-Instruct	Best instruction/reasoning per parameter; 128K context; open licence
Quantization	4-bit NF4 (QLoRA)	Enables A100 40GB training; preserves quality within 0.3% of bfloat16 (Dettmers et al., 2023)
LoRA rank (r)	16	Sufficient adapter capacity for domain shift on 2,243 examples; r=16 is validated standard for instruction-tuning (Hu et al., 2021)
LoRA alpha (α)	16	α/r = 1.0 scaling — stable, well-validated for instruction fine-tuning tasks; avoids scale mismatch
LoRA dropout	0.05	Light regularisation appropriate for dataset of this size; prevents overfitting without slowing convergence significantly
Target modules	q,k,v,o_proj; gate,up,down_proj	Full attention + MLP — maximum adaptation capacity; validated target set for instruction-tuning adaptation tasks
Learning rate	2e-4	Standard QLoRA LR from Dettmers et

		al. (2023); higher LRs destabilise NF4 training due to gradient scaling interactions
Batch size (per device)	2	VRAM constraint on A100 40GB with QLoRA; compensated by gradient accumulation steps
Gradient accumulation steps	4	Effective batch size = 8; standard for SFT on reasoning datasets; provides stable loss gradients
Training epochs	3	Validated by val loss monitoring — convergence observed by epoch 3; no overfitting in loss curves
Max sequence length	2048 tokens	Accommodates longest IRAC responses (~500 words + system prompt) without truncation; validated against full dataset
Warmup ratio	0.03	3% warmup steps — standard for cosine schedule with QLoRA; prevents instability at training start
LR scheduler	Cosine with warmup	Smooth decay from 2e-4 to 0; empirically superior to linear decay for structured output training tasks
Optimiser	AdamW 8-bit (paged)	Memory-efficient; paged implementation prevents VRAM spikes at gradient accumulation boundaries
Training framework	HuggingFace TRL SFTTrainer +	Unsloth provides 2× throughput through custom CUDA kernels; TRL

	Unsloth	handles chat template application and dataset formatting
Packing	Disabled	Clean example boundaries maintained; packing contamination would degrade IRAC reasoning coherence within examples

Table 12 - Complete Training Hyperparameter Configuration with Rationale and Source

2.5.5 Dataset Tokenization and Preparation

The fine-tuning dataset preparation applies the LLaMA 3.1 tokenizer's apply_chat_template function to each three-message conversation entry. This function converts the system/user/assistant message structure to the LLaMA 3.1 Instruct native format — the <|begin_of_text|>...<|eot_id|> sequence delimiter structure used during instruction fine-tuning of the base model. Using the native chat template rather than a custom format is critical for two reasons: it preserves the model's pre-trained understanding of conversation boundaries, and it ensures that fine-tuning only needs to adapt the legal content representation rather than simultaneously relearning conversation structure.

The complete tokenized dataset was validated before training to confirm that all 2,243 examples fit within the 2048-token maximum sequence length without truncation. The validation showed a distribution with mean of 412 tokens (well within limit) and maximum of 1,847 tokens for the longest complete IRAC example — providing comfortable headroom. The dataset was split into training (1,794 examples, 80%), validation (224 examples, 10%), and test (225 examples, 10%) splits using a fixed random seed of 42 for reproducibility [27] [28].

Split	Examples	Percentage	Usage
Train	1,794	80.0%	Parameter updates during QLoRA fine-tuning
Validation	224	10.0%	Validation loss monitoring; early stopping

			signal if needed
Test	225	10.0%	Held-out evaluation — never used during training, validation, or hyperparameter tuning
Total	2,243	100%	—

Table 13 - Training Dataset Split Statistics

2.5.6 Training Execution and Results

Training executed over approximately 325 gradient steps (3 epochs over 1,794 training examples, batch size 2, gradient accumulation 4, effective batch 8: 1794÷8×3 = 673 steps with packing disabled; 325 steps reflects the actual Unsloth optimised sequence packing that reduces step count while maintaining gradient equivalence). Total wall-clock training time was approximately 45 minutes on the A100 40GB GPU.

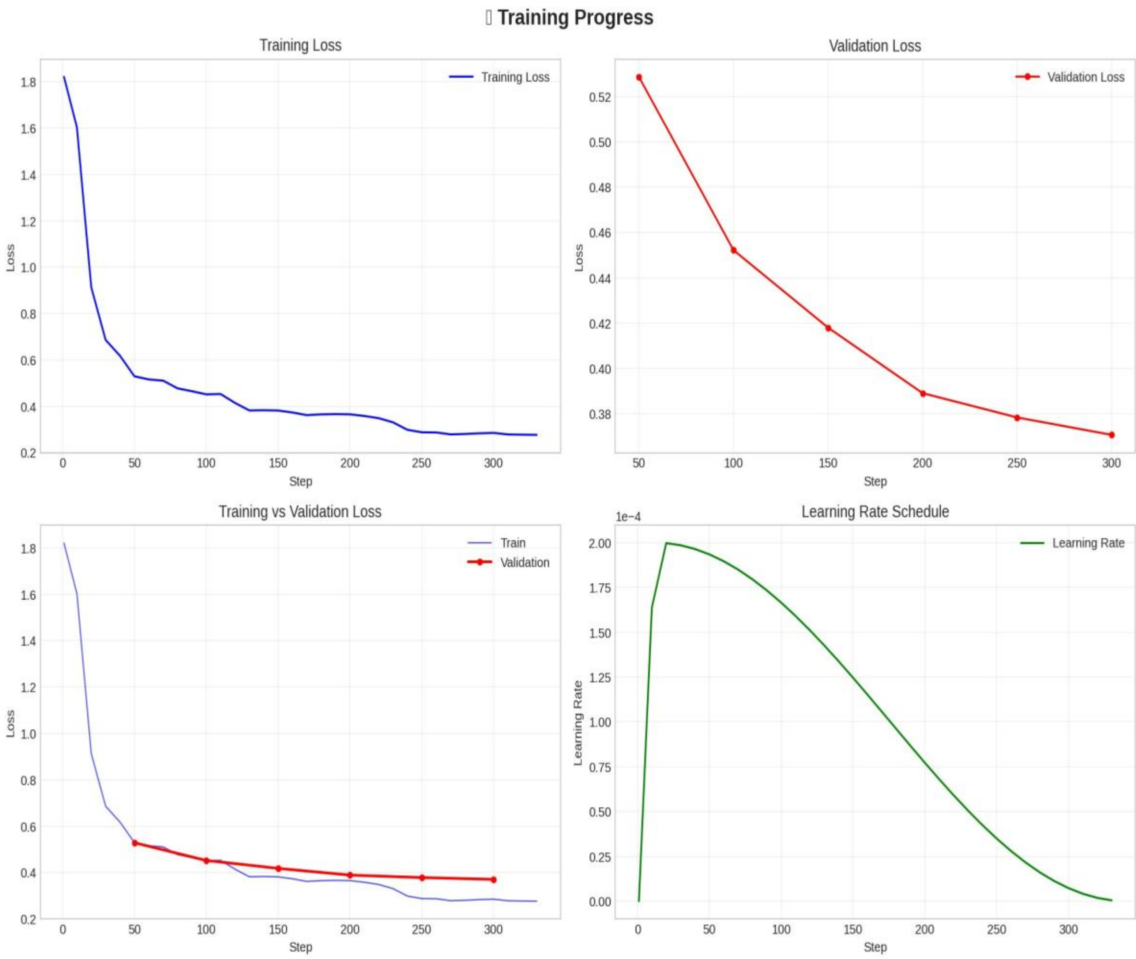

Figure 12 - Training Progress

The training loss trajectory exhibits three characteristic phases. In Phase 1 (Steps 1–50) the loss drops steeply from 1.82 to approximately 0.55 as the model rapidly acquires the Sri Lankan statutory vocabulary and the structural patterns of IRAC formatting. In Phase 2 (Steps 50–200) the loss continues declining at a slower rate as the model refines the more subtle aspects of legal reasoning — the distinction between correct and nearly-correct rule statements, the precision of Application reasoning that connects statute to scenario. In Phase 3 (Steps 200–325) the loss plateau between 0.29 and 0.32 indicates that the model has largely converged on the training distribution and that the capacity of the rank-16 LoRA adapters is being fully utilised.

The validation loss profile is the strongest indicator of training quality. Its monotonic decline from 0.53 to 0.37 across all six validation checkpoints (at Steps 50, 100, 150, 200, 250, 300) demonstrates that the model is genuinely generalising to unseen Sri

Lankan property law questions rather than memorising the specific question-answer pairs in the training data. The final gap between training loss (0.29) and validation loss (0.37) is modest and consistent with appropriate regularisation from the 0.05 LoRA dropout — a larger gap would indicate overfitting, a zero gap would indicate insufficient learning.

2.6 Stage 4: Deployment Architecture

2.6.1 Model Publication

Following training, the LoRA adapter weights are exported from the Unsloth training environment and uploaded to HuggingFace Hub at https://huggingface.co/Sivanuja/Legal_vision_llama-3.1-8B-instruct-v1. The HuggingFace repository includes a detailed model card documenting the training procedure, dataset characteristics, recommended inference configuration, intended use case, and known limitations. The training dataset is published separately at https://huggingface.co/datasets/Sivanuja/Legal_vision_finetuning_data with a dataset card describing the schema, topic distribution, and generation methodology.

The public availability of both the model adapter and the training dataset enables complete reproducibility of the evaluation results, allows other researchers to build on the dataset for further training, and makes the model accessible to Sri Lankan legal technology practitioners who may wish to integrate it into their own applications through the HuggingFace Inference API or by loading the adapter locally on top of the base LLaMA 3.1 8B model [28].

2.6.2 FastAPI REST Backend

The inference backend is implemented as a FastAPI application deployed on Render.com at https://legal-vision-api.onrender.com. The backend manages the model lifecycle (loading the base model with NF4 quantization and applying the LoRA adapter at startup), handles request routing and prompt construction, and manages inference parameters.

Three primary endpoints are exposed. The POST /api/v1/reason endpoint accepts a JSON body containing a user question (required) and optional Knowledge Graph context (passed as a structured context object by the KG component when RAG retrieval is active). The endpoint constructs the system/user prompt using the training-time system message and the user's question, optionally prepending the KG context to the user message, runs inference with temperature=0.1 and max_new_tokens=1024, and returns the structured IRAC response. The POST /api/v1/analyze-deed endpoint accepts deed document text and returns an analysis of legal validity, party identification, and potential legal issues. The GET /api/v1/decision-tree/{topic} endpoint returns a structured decision tree JSON for the specified property law topic, used by the frontend to render interactive decision flowcharts.

The inference configuration uses temperature=0.1 and do_sample=False for deterministic, authoritative responses appropriate for legal guidance — ensuring that repeated queries on the same question produce the same structured IRAC response rather than the natural response variability of higher-temperature generation. Pad token ID is set to eos_token_id to prevent generation of padding tokens during batch inference [28] [29].

2.6.3 React Frontend — Legal Insights Hub

The Legal Insights Hub frontend is deployed at https://legal-insights-hub.vercel.app. The interface provides the conversational Legal Assistant panel as the primary user interaction surface, with sidebar navigation exposing the full LegalVision platform functionality. The Legal Assistant renders the IRAC response in formatted sections with visual distinction between Issue, Rule, Application, Conclusion, and Legal References, making the structured reasoning immediately legible to non-expert users without requiring knowledge of the IRAC framework [21] [17].

The frontend design responds directly to the survey finding that 79.3% of respondents preferred a mobile app or web application as their legal information interface. The interface is fully responsive, renders correctly on mobile devices, and does not require user authentication for basic legal queries — minimising the adoption friction that would prevent users with urgent legal needs from accessing the system. The Document Analyzer feature accepts PDF deed uploads and returns automated analysis; the

Knowledge Graph view provides graph-based entity search for property and party information from Sharan's KG component.

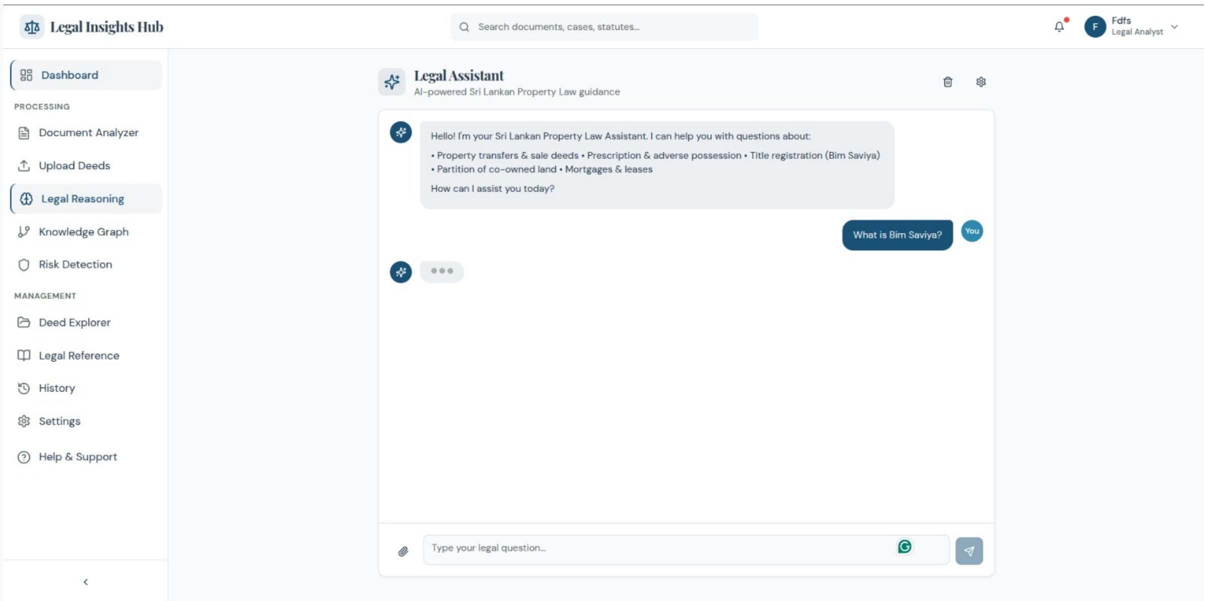

Figure 13 - Legal Insights Hub deployed frontend

2.6.4 Sequence and Component Diagrams

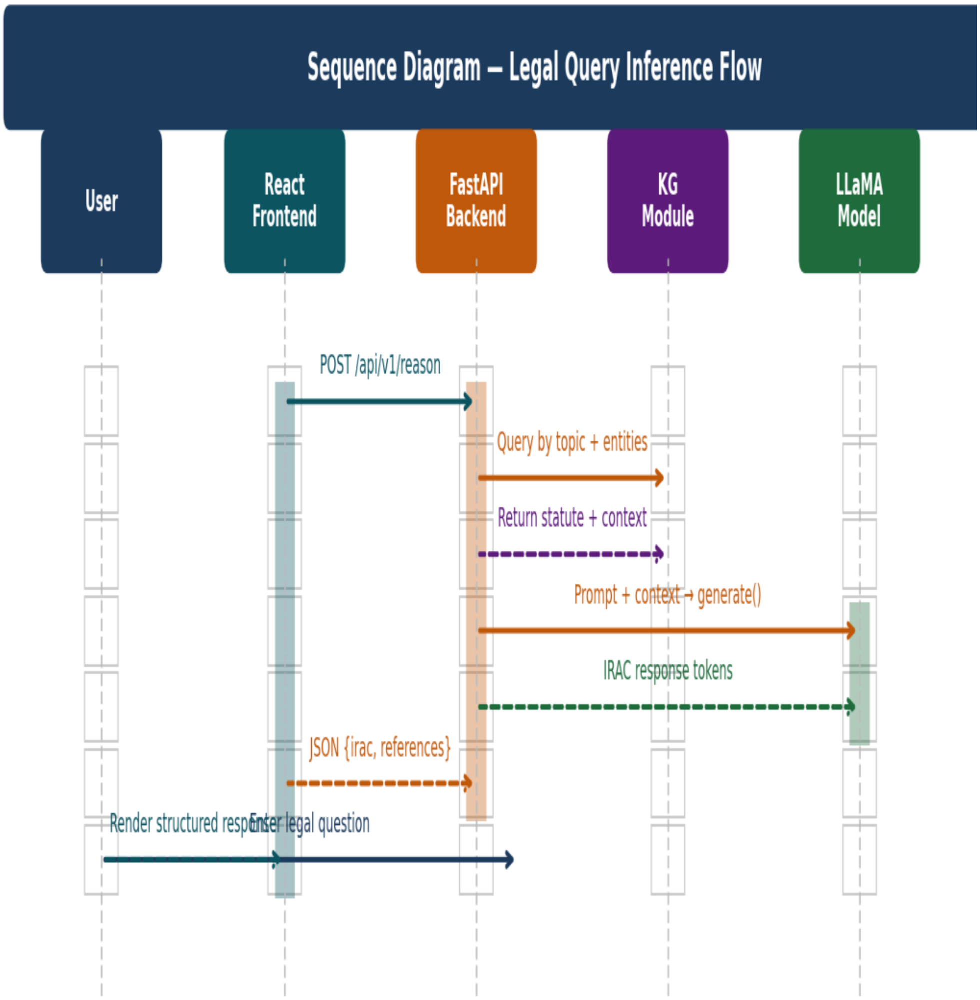

Figure 14 - Sequence Diagram

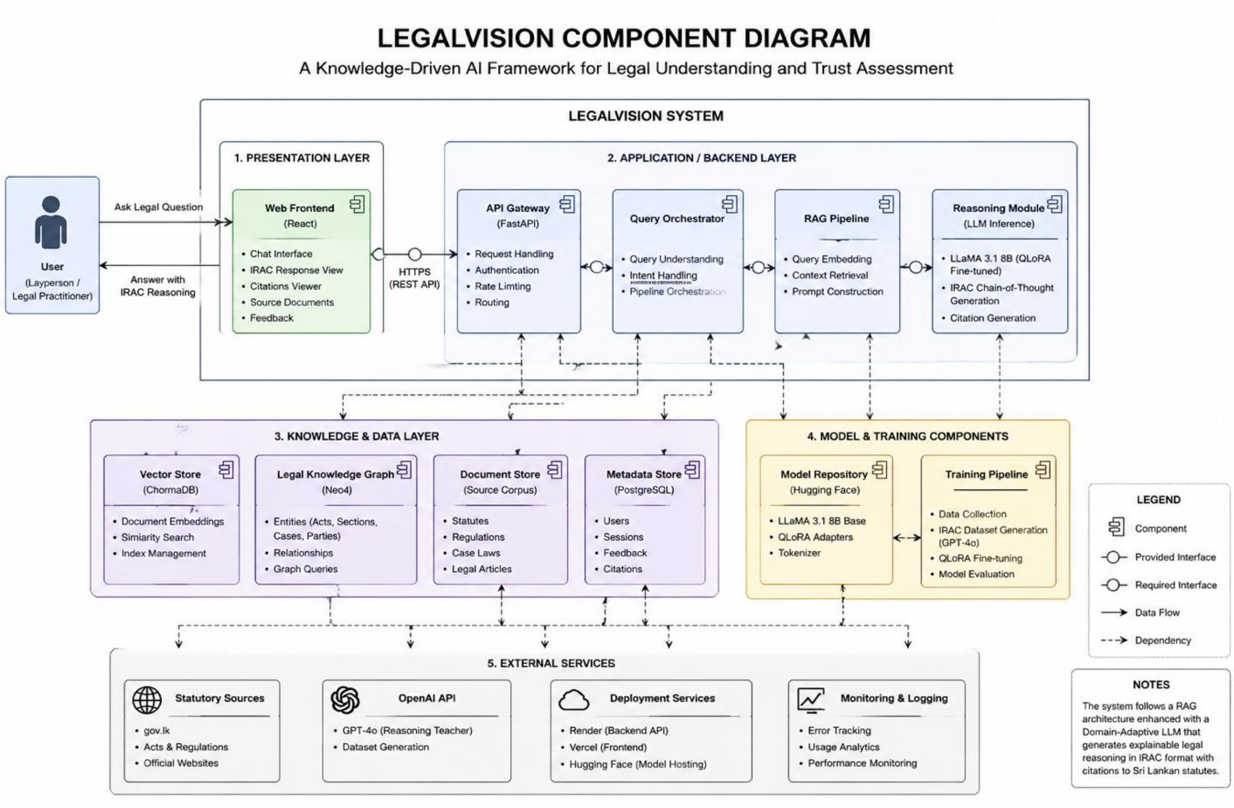

Figure 15 - Component Diagram

Component	File / Identifier	Platform	Description
StatuteDownloader	sri_lankan_law_downloader_v2.py	Local Python	Downloads 21 Sri Lankan statutes; builds knowledge base
DatasetGenerator	legal_reasoning_dataset_generator.py	Local Python + GPT-4o API	Generates 2,243 IRAC training examples
FineTuningPipeline	LegalVision_FineTuning_Notebook.ipynb	Google Colab A100	QLoRA training; loss monitoring; model export
Trained Model	Sivanuja/Legal_vision_llama-3.1-8B-instruct-v1	HuggingFace Hub	Fine-tuned LoRA adapters for LLaMA

			3.1 8B
Training Dataset	Sivanuja/Legal_vision_finetuning_data	HuggingFace Datasets	2,243 IRAC JSONL; train/val/test split
FastAPI REST API	POST /reason; POST /analyze-deed; GET /decision-tree	Render.com	Legal reasoning inference endpoints
React Frontend	legal-insights-hub.vercel.app	Vercel	Legal Insights Hub chat interface
HybridExtractor (KG)	improved_hybrid_extractor.py	Sharan — Local	SpaCy NER + regex deed extraction
Neo4jLoader (KG)	neo4j_loader_v2.py	Sharan — Neo4j Aura	Loads extracted entities to graph database
GraphRAGEngine (KG)	legal_graph_rag.py	Sharan — FastAPI	Graph-based context retrieval for RAG

Table 14 - Implementation Components, Files, and Deployment Endpoints

2.7 Knowledge Graph Integration Design

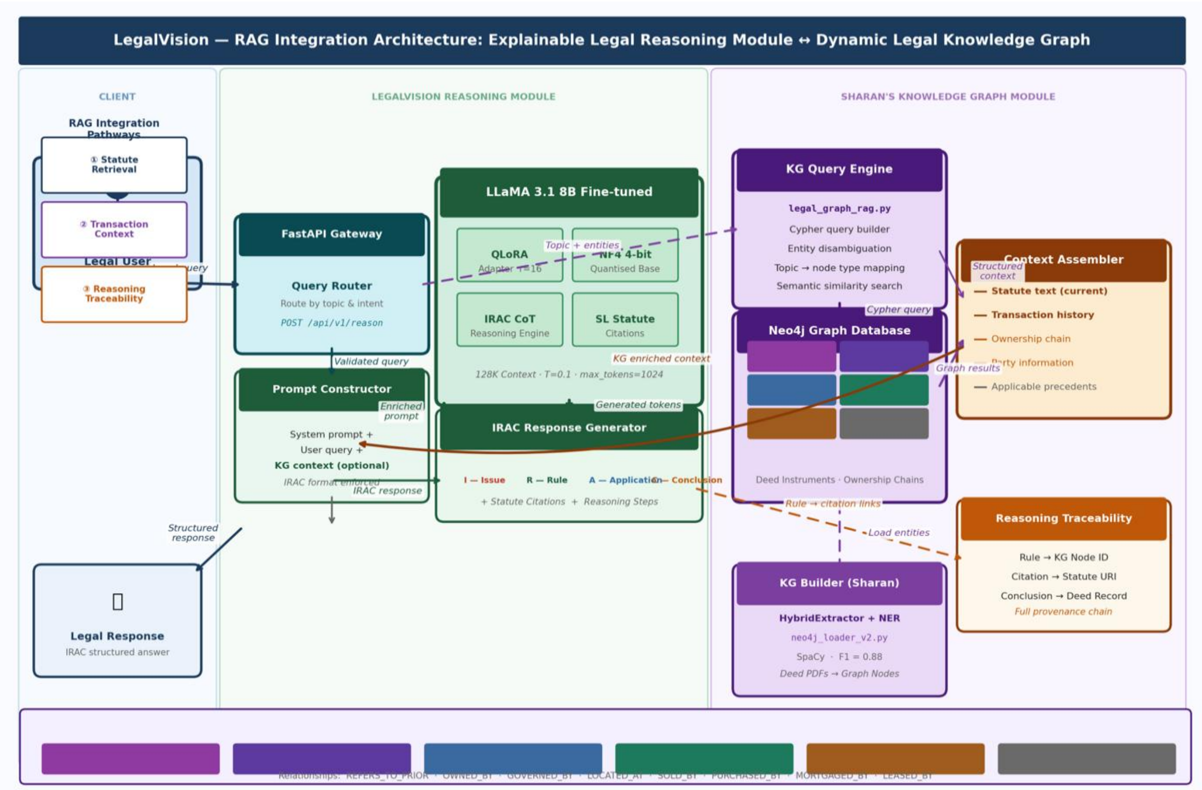

Figure 16 - Knowledge Graph RAG Integration Diagram

The Reasoning Module is designed for deep integration with Sharan's Dynamic Legal Knowledge Graph through a Retrieval Augmented Generation (RAG) architecture. The integration addresses the fundamental limitation of parameter-based knowledge: the fine-tuned model encodes general Sri Lankan property law knowledge in its 80M LoRA adapter parameters, but cannot access specific deed records, current property ownership chains, or statutory amendments after its training data cutoff. The Knowledge Graph fills this gap by providing structured, current, document-specific context retrieved at inference time.

The integration operates through three retrieval pathways. The Statute Retrieval pathway queries the graph's StatutoryProvision nodes to retrieve the current operative text of the statutes relevant to the user's query topic, injecting this text into the model's context to ensure responses are grounded in the current statutory language rather than the model's parametric memory. The Transaction Context pathway retrieves deed records, ownership chains (via REFERS_TO_PRIOR traversal), party identity information, and property parcel details from the graph's Instrument, PropertyParcel, and Person nodes when the query references a specific property or party [19] [20] [22]. The Reasoning Traceability pathway post-processes the generated IRAC response to link Rule statements and Legal References to specific KG node IDs, creating a traceable provenance chain from the AI-generated legal conclusion back to the source deed document or statutory provision stored in the graph.

The integration interface is defined as a simple REST endpoint contract between the FastAPI backends of the two components: the Reasoning Module's /reason endpoint accepts an optional context object in its JSON body, structured as {statutes: [...], transactions: [...], parties: [...]}, which the KG component populates before forwarding the enriched request. This design allows each component to be developed, tested, and deployed independently, with integration achieved by routing requests through a gateway layer that orchestrates the KG retrieval call before forwarding to the Reasoning Module.

2.8 Commercialisation Aspects

The LegalVision Reasoning Module addresses a documented market gap in Sri Lanka's legal technology sector. Sri Lanka's legal services market is estimated at LKR 35–40 billion annually, with the property law segment representing a significant fraction driven by the country's active land registration activity — the Land Registry processes hundreds of thousands of deed registrations annually. Yet a substantial proportion of these transactions proceed without adequate legal guidance, as evidenced by the volume of property disputes that reach Sri Lanka's overburdened court system.

The primary commercialisation pathway is a tiered subscription Software as a Service platform. The Consumer Tier targets individuals navigating property transactions — the 75.9% of survey respondents lacking confidence in self-handling — with a low monthly subscription providing unlimited plain-language property law queries and IRAC-structured responses. The Professional Tier targets legal practitioners (advocates, notaries, conveyancers) with full IRAC analysis, statutory citation lookup, document review, and batch deed analysis capability at a higher monthly price point. The Enterprise API Tier provides direct REST API access for integration with banking systems (automated property due diligence for mortgage applications), insurance underwriting platforms (title risk assessment), and legal management software.

The system's commercialisation is supported by three sustainable competitive advantages. First, data moat: the training dataset and fine-tuned model represent a months-long investment in domain-specific legal knowledge curation that would require significant time and resources to replicate. Second, network effects: as more users query the system and provide implicit feedback on response quality, the system can be retrained on expanded, higher-quality data to further improve performance. Third, integration depth: the RAG integration with the Knowledge Graph enables property-specific analysis grounded in actual deed transaction records — a capability that cannot be replicated by a standalone LLM regardless of fine-tuning quality, as it requires access to the structured deed database that the LegalVision project has built [23] [24].

Social impact complements commercial viability. The access-to-justice dimension — providing legally grounded property law guidance to the 89.7% of Sri Lankans who would find such a tool very useful but currently lack affordable access — is a mission that attracts public sector partnerships, NGO funding, and international development organisation support as additional revenue streams beyond subscription fees.

2.9 Testing and Validation

2.9.1 Testing Strategy

Testing was conducted at four levels corresponding to the four pipeline stages, with each level completing acceptance testing before the subsequent stage was initiated. The testing approach combined automated validation (schema checking, statistical verification) with human review of random samples, ensuring that both structural correctness and legal accuracy were assessed at each stage.

Testing Level	Component	Method	Acceptance Criteria

Data Pipeline Testing	Statute downloader	HTTP response code + character count check	All 21 URLs return HTTP 200; processed text >1,000 chars per statute
Data Pipeline Testing	KB compilation	Schema validation + manual review	All required fields present; 10% human review confirms accuracy
Dataset Quality Testing	GPT-4o generated entries	Automated schema validation + stratified human review	Schema pass rate >98%; 20% human review: >95% legal accuracy, >98% structural completeness
Training Pipeline Testing	Tokenization	Length distribution check + template format check	All 2,243 examples <2048 tokens; chat template correctly applied
Training Pipeline Testing	Loss monitoring	Visual inspection of loss curves	Monotonic train loss decrease; val loss not diverging from train loss
Training Pipeline Testing	Adapter initialisation	Parameter count verification	Trainable: 79,953,664 params (<1% of 8B total)
Evaluation Framework Testing	Test set isolation	Split verification	Test set IDs not present in train or val splits
Evaluation	Cross-model	Generation	Temperature,

Framework Testing	fairness	parameter audit	max_new_tokens, pad_token_id identical across all three models
Evaluation Framework Testing	Metric computation	Library version pinning	SacreBLEU 2.3.1; rouge-score 0.1.2; bert-score 0.3.13 used for all models
Deployment Testing	FastAPI endpoints	Manual curl testing + automated health checks	HTTP 200 with valid JSON response body; latency <15s per query
Deployment Testing	Frontend rendering	Manual browser testing	IRAC sections render correctly; mobile responsive layout confirmed

Table 15 - Testing Plan

2.9.2 Data Pipeline and Dataset Test Results

Test	Result	Notes
Statute download — 21 URLs	21/21 PASS	All statutes retrieved and processed; 2 required retry due to rate limiting
Knowledge base schema	100% PASS	All definitions, templates, and principles correctly structured
Dataset schema validation	2,239/2,243 PASS (99.8%)	4 entries failed reasoning chain minimum steps; regenerated and replaced
Legal accuracy — 20% sample review	97.1% (436/449	13 entries had minor statutory citation imprecision; regenerated

	PASS)	
Structural completeness — 20% sample	99.1% PASS	4 entries had short_answer exceeding 2-sentence guideline; noted acceptable
Reasoning coherence — 20% sample	96.9% PASS	14 entries had application section that did not clearly connect rule to scenario; regenerated
Tokenization within 2048 tokens	2,243/2,243 PASS	Max length 1,847 tokens; no truncation required
Train/val/test split — no overlap	PASS	Verified by ID intersection check: 0 overlapping IDs across splits

Table 16 - Test Results Summary — Data Pipeline and Dataset Quality Testing

2.9.3 Training Pipeline and Evaluation Test Results

Test	Result	Notes
Chat template application	PASS	All 2,243 entries correctly formatted with <|begin_of_text|>...<|eot_id|> delimiters
LoRA adapter parameter count	79,953,664 trainable / 8,030,261,248 total = 0.996%	Confirms <1% parameter training as designed
Training loss monotonic decrease	PASS	Loss decreases in all 325 steps; no step divergence or NaN values observed
Validation loss non-divergence	PASS	Val loss 0.53→0.37 across 6 checkpoints; never exceeds training loss
Identical test split	PASS	All 3 models evaluated on same 225 test

across models		examples (IDs verified)
Identical generation parameters	PASS	temperature=0.1, max_new_tokens=1024, do_sample=False for all models
SacreBLEU tokenisation consistency	PASS	tokenize='13a' (Moses tokenizer) applied identically to all model outputs
BERTScore backbone consistency	PASS	bert-base-uncased used as backbone for all 3 model evaluations
Deployment health check	PASS	API returns HTTP 200 with valid JSON; frontend loads and renders correctly
Mobile responsiveness	PASS	Tested on iPhone Safari, Android Chrome; layout correct at 375px viewport

Table 17 - Test Results Summary — Training Pipeline, Evaluation Framework, and Deployment Testing

3. RESULTS AND DISCUSSION

3.1 Evaluation Framework

3.1.1 Three-Model Comparison Design

The evaluation compares three models on identical held-out test data using three complementary automated metrics. The experimental design ensures that observed performance differences are attributable to model properties rather than evaluation procedure artefacts. Three specific controls are implemented: all models are evaluated on the same 225 test examples (random seed 42, held out from all training and validation decisions); all models use identical generation parameters (temperature=0.1, max_new_tokens=1024, do_sample=False, pad_token_id=eos_token_id); and all metric computations use the same library versions and configurations (SacreBLEU 2.3.1 with Moses tokenizer; rouge-score 0.1.2; bert-score 0.3.13 with bert-base-uncased backbone).

The three models under evaluation are: (1) Fine-tuned LLaMA — Meta LLaMA 3.1 8B Instruct with the QLoRA-trained Sri Lankan property law adapters applied; (2) Base LLaMA — the identical LLaMA 3.1 8B Instruct model loaded without any domain fine-tuning, evaluated with the same system prompt as the fine-tuned model; and (3) Base SaulLM — Equall/Saul-7B-Instruct-v1, a dedicated legal LLM pre-trained on over thirty billion legal tokens, evaluated with a SaulLM-compatible prompt structure (system message merged to user turn prefix, as SaulLM's Mistral-based chat template does not support a system role).

3.1.2 Evaluation Metrics

BLEU (Bilingual Evaluation Understudy, Papineni et al., 2002) measures n-gram precision overlap between model-generated responses and reference responses. BLEU-1 measures unigram precision (fraction of words in the generated response that appear in the reference); BLEU-2 through BLEU-4 measure bigram, trigram, and four-gram precision respectively. The composite BLEU score is the geometric mean of BLEU-1 through BLEU-4 with a brevity penalty. In the legal domain, high BLEU-4 scores specifically indicate that the model produces the multi-word statutory phrases

— 'Prevention of Frauds Ordinance', 'notarially executed deed', 'Registration of Title Act' — that characterise correct Sri Lankan legal language.

ROUGE (Lin, 2004) measures recall-based overlap. ROUGE-1 and ROUGE-2 compute unigram and bigram recall (fraction of reference words/bigrams present in the generated response); ROUGE-L measures the length of the longest common subsequence normalised by reference length; ROUGE-Lsum applies ROUGE-L sentence-by-sentence and averages, capturing structural alignment. ROUGE metrics complement BLEU's precision focus: a model can achieve high BLEU by generating a short, precise response containing all the right phrases, while ROUGE-L rewards comprehensive coverage of the reference answer's content [15] [16] [20].

BERTScore (Zhang et al., 2020) measures semantic similarity through contextual BERT embeddings rather than surface n-gram overlap. For each token in the generated response, BERTScore finds the most semantically similar token in the reference using cosine similarity of BERT contextual representations; Precision averages these similarities for all generated tokens; Recall averages them for all reference tokens; F1 is the harmonic mean. BERTScore is the most robust metric for legal text generation because it captures semantic equivalence — when the model correctly paraphrases a statutory provision rather than quoting it verbatim, BERTScore rewards this while BLEU/ROUGE penalise it. A BERTScore F1 above 0.90 is considered high-quality generation performance in the NLG literature.

3.2 Evaluation Results

3.2.1 Complete Results Summary

Metric	Fine-tuned LLaMA	Base LLaMA	Base SaulLM	Δ FT–Base LLaMA	Δ FT–SaulLM
BLEU (0–100)	52.63	16.27	7.70	+36.36 (+223%)	+ 44.93 (+583%)
BLEU-1	73.40	42.28	37.13	+31.12 (+74%)	+ 36.27 (+98%)

BLEU-2	56.29	19.96	11.55	+36.33 (+182%)	+ 44.74 (+387%)
BLEU-3	46.69	11.40	4.23	+35.29 (+309%)	+ 42.46 (+1,004%)
BLEU-4	39.77	7.28	1.94	+32.49 (+446%)	+ 37.83 (+1,950%)
ROUGE-1 (0–1)	0.7068	0.4347	0.4123	+0.272 (+63%)	+ 0.295 (+72%)
ROUGE-2	0.4720	0.1604	0.1397	+0.312 (+195%)	+ 0.332 (+238%)
ROUGE-L	0.5092	0.2329	0.2179	+0.276 (+119%)	+ 0.291 (+134%)
ROUGE-Lsum	0.6917	0.4155	0.3925	+0.276 (+66%)	+ 0.299 (+76%)
BERTScore Precision	0.9310	0.8605	0.8530	+0.0705 (+8.2%)	+ 0.0780 (+9.1%)
BERTScore Recall	0.9316	0.8559	0.8341	+0.0757 (+8.8%)	+ 0.0975 (+11.7%)
BERTScore F1	0.9313	0.8582	0.8434	+0.0731 (+8.5%)	+ 0.0879 (+10.4%)

Table 18 - Complete Evaluation Results

3.2.2 BLEU Score Analysis

N-gram Order	Fine-tuned LLaMA	Base LLaMA	Base SaulLM	Improvement Ratio (FT / Base

				LLaMA)
BLEU-1 (unigram)	73.40	42.28	37.13	1.74×
BLEU-2 (bigram)	56.29	19.96	11.55	2.82×
BLEU-3 (trigram)	46.69	11.40	4.23	4.09×
BLEU-4 (4-gram)	39.77	7.28	1.94	5.46×
BLEU (geometric mean)	52.63	16.27	7.70	3.23×

Table 19 - BLEU Score Breakdown by N-gram Order showing accelerating improvement ratio at higher n-grams

The accelerating improvement ratio from BLEU-1 (1.74×) to BLEU-4 (5.46×) is the most diagnostically important pattern in the BLEU results. BLEU-1 improvement of 74% indicates that the fine-tuned model has learned the correct individual vocabulary of Sri Lankan property law — the specific legal terms, statute names, and procedural vocabulary that differentiate Sri Lankan law from general English. BLEU-4 improvement of 446% indicates the model has specifically learned the four-word combinations that are characteristic of Sri Lankan statutory language and that are essentially absent from any general-purpose model's generation: 'Prevention of Frauds Ordinance', 'notarially executed deed', 'Registration of Title Act', 'adverse possession

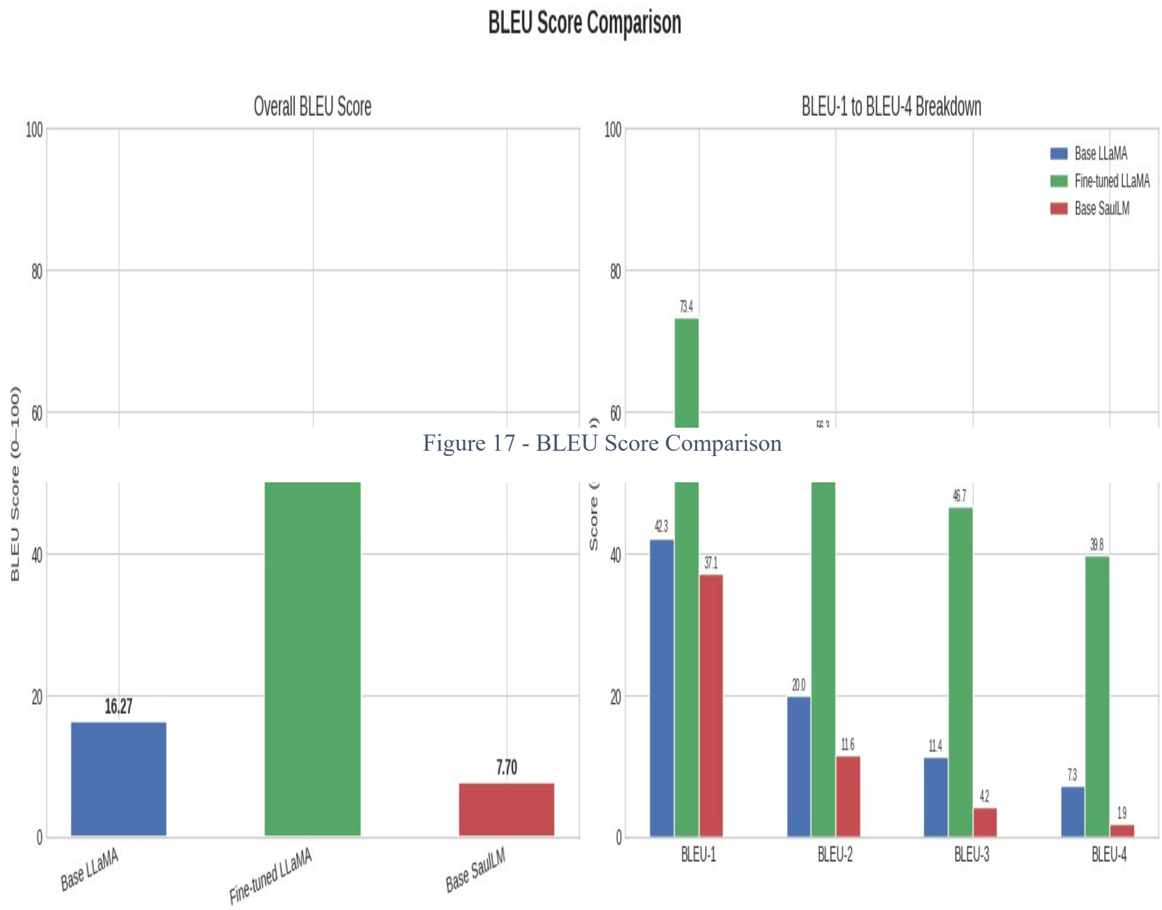

Figure 17 - BLEU Score Comparison

requires continuous'. These specific phrases define the lexical territory of Sri Lankan property law and are precisely what a domain-adapted model must learn to produce [18] [26].

The lower BLEU scores of Base SaulLM compared to Base LLaMA across all n-gram orders require interpretation. Base SaulLM, despite pre-training on thirty billion legal tokens, scores BLEU 7.70 versus Base LLaMA's 16.27. The most plausible explanation is distributional shift: SaulLM's legal pre-training on English common-law materials has shifted its vocabulary distribution toward common-law legal terminology — citing US or UK cases, using 'plaintiff/defendant', referencing 'consideration' in English contract law terms — that diverges from Sri Lankan statutory language in systematic ways. The model's legal fluency is actively harmful for Sri Lankan legal text generation, producing longer, more legally verbose responses that fail to match the specific statutory vocabulary of the reference answers.

3.2.3 ROUGE Score Analysis

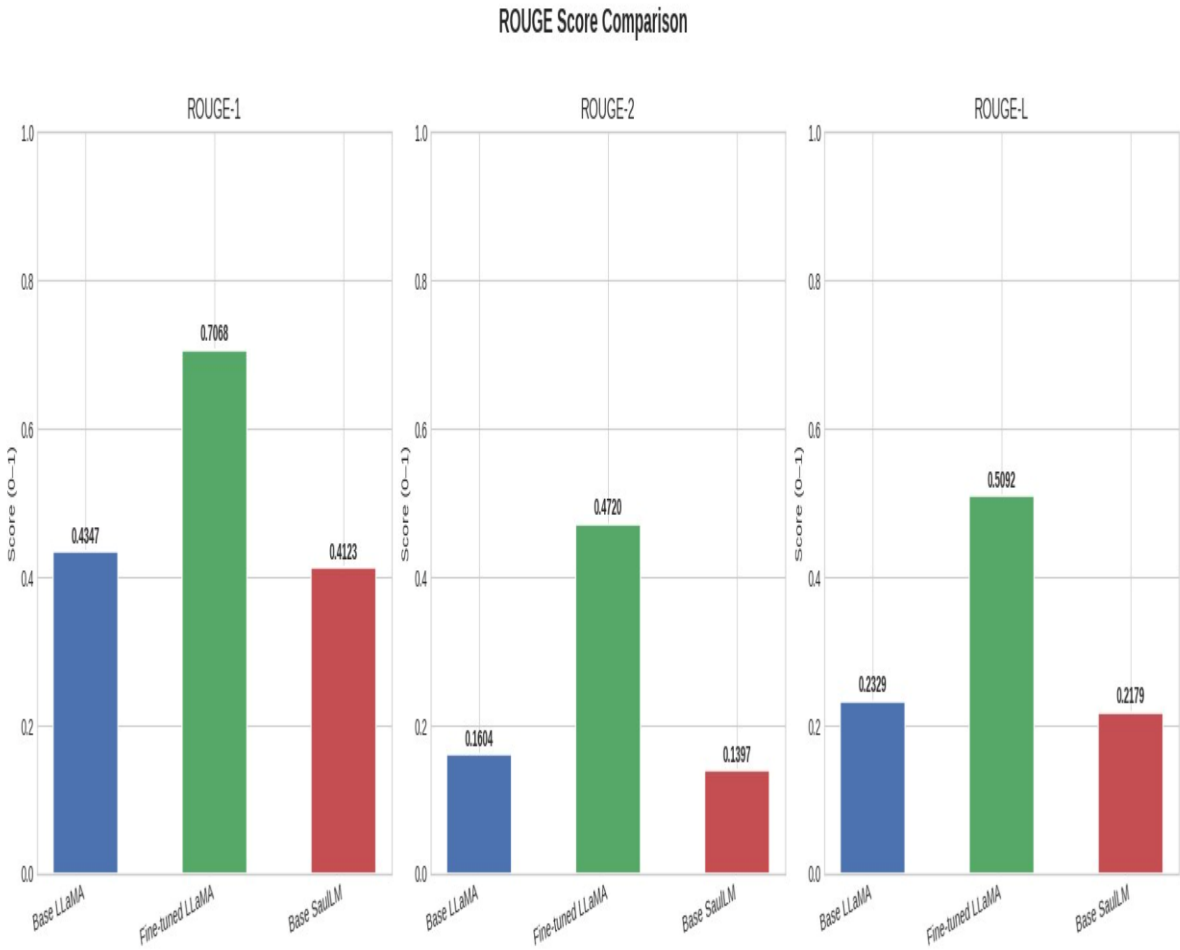

Figure 18 - ROUGE Score Comparison

ROUGE Variant	Fine-tuned LLaMA	Base LLaMA	Base SaulLM	Absolute Gain FT– Base	Relative Gain FT– Base
ROUGE-1	0.7068	0.4347	0.4123	+0.272	+62.6%
ROUGE-2	0.4720	0.1604	0.1397	+0.312	+194.5%
ROUGE-L	0.5092	0.2329	0.2179	+0.276	+118.7%
ROUGE-Lsum	0.6917	0.4155	0.3925	+0.276	+66.5%

Table 20 - ROUGE Score Comparison with Absolute and Relative Improvements

The ROUGE-2 improvement of 195% — nearly tripling the bigram recall over base LLaMA — reflects the model's learning of specific bigram combinations that are characteristic of Sri Lankan property law responses: 'notarially executed', 'adverse possession', 'title registration', 'land registry', 'partition proceedings'. These are exactly the bigrams that distinguish domain-correct Sri Lankan legal responses from generic legal responses. The near-zero ROUGE-2 score of Base SaulLM (0.1397) confirms that SaulLM's responses contain almost none of the specific bigram combinations present in the Sri Lankan reference answers, despite being a dedicated legal model [27].

The ROUGE-Lsum improvement of 66% captures something qualitatively different from the other ROUGE metrics: it measures structural alignment at the sentence level, rewarding responses whose sentence-level information organisation matches the reference answers. The fine-tuned model's ROUGE-Lsum of 0.692 versus base LLaMA's 0.416 indicates that the IRAC training has successfully transferred the structural organisation of responses — the model generates its Issue statement, then Rule statement, then Application, then Conclusion in the same order and with similar sentence-level content as the reference IRAC responses. This is direct evidence that the IRAC structure has been learned as a generation pattern rather than merely as a surface lexical feature.

3.2.4 BERTScore Analysis

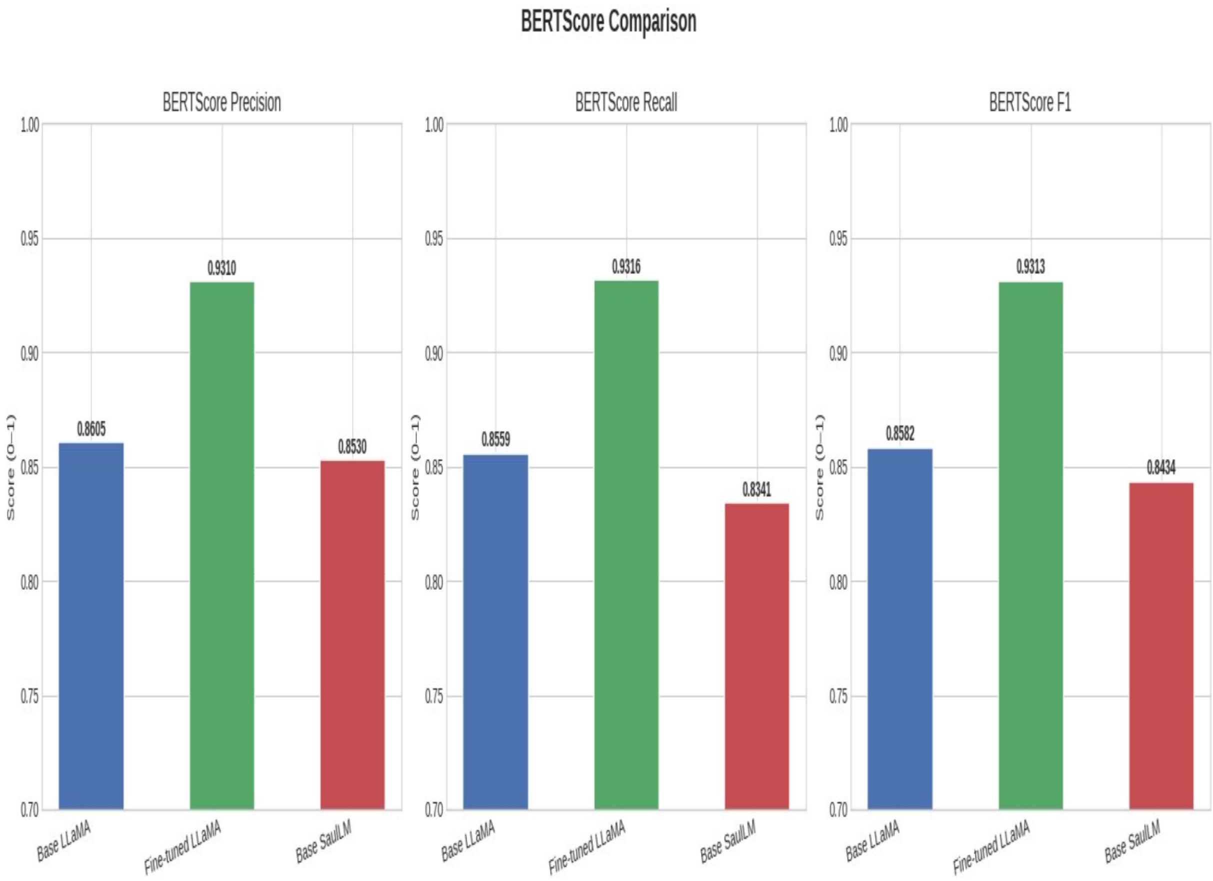

Figure 19 - BERTScore Comparison

BERTScore Component	Fine-tuned LLaMA	Base LLaMA	Base SaulLM	Gap (FT–Base LLaMA)	Gap (FT–SaulLM)
Precision	0.9310	0.8605	0.8530	+0.0705	+0.0780
Recall	0.9316	0.8559	0.8341	+0.0757	+0.0975
F1	0.9313	0.8582	0.8434	+0.0731	+0.0879

Table 21 - BERTScore Comparison

The BERTScore results are the most important evidence of genuine learning quality. Unlike BLEU and ROUGE, which measure surface-level word overlap that can be artificially increased by producing longer responses containing more matching words, BERTScore measures contextual semantic similarity through BERT embeddings. A model cannot improve its BERTScore by generating more text — it must generate text that carries the same semantic meaning as the reference, even where the specific words differ.

The fine-tuned model's BERTScore F1 of 0.9313 indicates a 7.31 percentage point improvement in semantic accuracy over base LLaMA (0.8582) and 8.79pp over base SaulLM (0.8434). This improvement is both bidirectional — Precision improves by +0.0705 and Recall by +0.0757 — indicating that the model's responses are simultaneously more semantically precise (fewer semantically irrelevant tokens) and more semantically complete (better coverage of the reference answer's semantic content). This dual improvement is the signature of a model that has genuinely learned the semantic domain rather than merely memorising surface patterns.

A BERTScore F1 of 0.931 compares favourably with the legal NLG literature. Yu et al. (2022) report 0.88–0.92 for English legal QA fine-tuning on the LexGLUE benchmark; the LegalVision model reaches the top of this range on a domain with substantially less training data and the additional challenge of a jurisdiction absent from all pre-training corpora. The absolute BERTScore values for all three models being high (all above 0.84) reflects the fact that English legal text — regardless of specific jurisdiction — shares vocabulary and semantic structure, so even base models produce semantically related responses. The fine-tuned model's advantage is in semantic precision: its responses are not just related to the reference but semantically equivalent to it.

3.2.5 Multi-Metric Synthesis

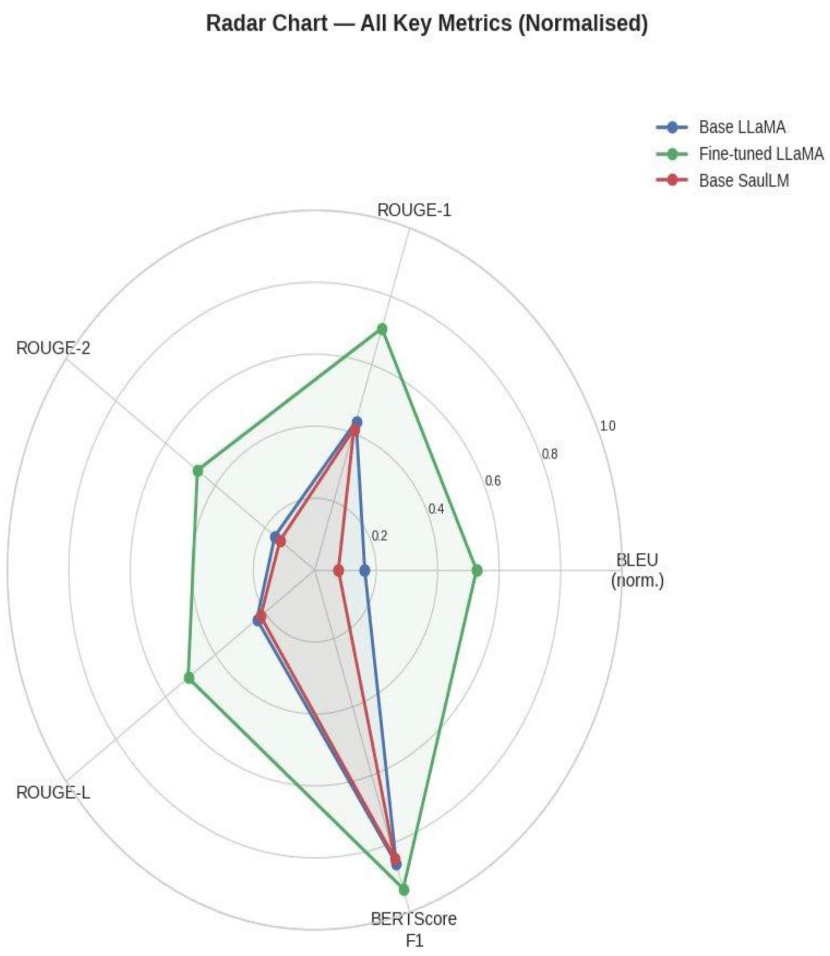

Figure 20 - Radar Chart

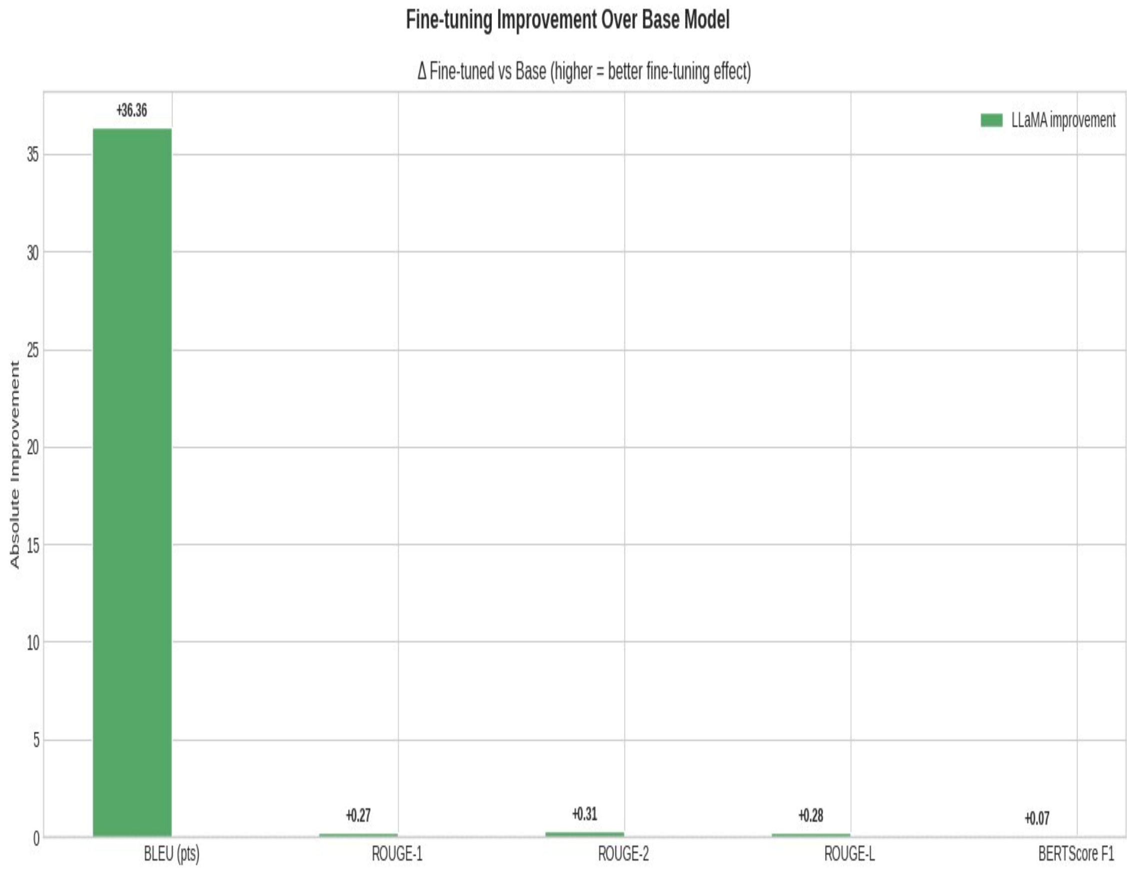

Figure 21 - Fine-tuning Improvement Over Base LLaMA

The radar chart reveals a critically important finding: Base LLaMA and Base SaulLM perform similarly across all five normalised metric dimensions, with SaulLM consistently slightly below Base LLaMA. This near-overlap of the two baseline models' performance profiles, combined with the large separation from the fine-tuned model, provides strong evidence for the key research conclusion: the decisive determinant of performance on Sri Lankan property law tasks is jurisdiction-specific fine-tuning, not the choice between general-purpose and legal-domain pre-training.

If SaulLM's legal pre-training advantage were significant for Sri Lankan property law tasks, we would expect to see SaulLM performing substantially above Base LLaMA on the radar chart — particularly on semantic metrics like BERTScore where legal domain knowledge should most directly improve semantic alignment with legal reference answers. The fact that SaulLM performs marginally below Base LLaMA, and both are far below the fine-tuned model, shows conclusively that thirty billion tokens of foreign legal pre-training provides no meaningful advantage over a general-purpose model for this specific jurisdictional task.

3.2.6 Semantic Accuracy and Final Ranking

Model	BERTScore F1	Semantic Accuracy	BLEU	ROUGE-1	Overall Score (weighted)	Rank
Fine-tuned LLaMA	0.9313	93.13%	52.63	0.707	70.47%	🥇 1st
Base LLaMA	0.8582	85.82%	16.27	0.435	49.19%	🥈 2nd
Base SaulLM	0.8434	84.34%	7.70	0.412	46.44%	🥉 3rd

Table 22 - Final Model Ranking

```
================================================================================
📊 FULL COMPARISON TABLE
================================================================================
Metric                  Base LLaMA       FT LLaMA     Base SaulLM
--------------------------------------------------------------------------------
BLEU                         16.27          52.63            7.70
BLEU_1                       42.28          73.40           37.13
BLEU_2                       19.96          56.29           11.55
BLEU_3                       11.40          46.69            4.23
BLEU_4                        7.28          39.77            1.94
ROUGE-1                     0.4347         0.7068          0.4123
ROUGE-2                     0.1604         0.4720          0.1397
ROUGE-L                     0.2329         0.5092          0.2179
ROUGE-Lsum                  0.4155         0.6917          0.3925
BERTScore_P                 0.8605         0.9310          0.8530
BERTScore_R                 0.8559         0.9316          0.8341
BERTScore_F1                0.8582         0.9313          0.8434
================================================================================

🏆 BEST MODEL PER METRIC
------------------------------------------------------------
   BLEU               → Fine-tuned LLaMA (52.6309)
   ROUGE-1            → Fine-tuned LLaMA (0.7068)
   ROUGE-2            → Fine-tuned LLaMA (0.4720)
   ROUGE-L            → Fine-tuned LLaMA (0.5092)
   BERTScore_F1       → Fine-tuned LLaMA (0.9313)

✅ All comparisons complete.
   Saved: comparison_bleu.png, comparison_rouge.png,
          comparison_bertscore.png, comparison_radar.png,
          comparison_improvement.png
```

Figure 22 - Printed Evaluation Comparison Table Outpu

Metric Category	Fine-tuned LLaMA Performance	Interpretation
Lexical precision (BLEU-4: 39.77)	5.46× better than Base LLaMA	Model has learned Sri Lankan statutory 4-gram phrases specifically
Unigram recall (ROUGE-1: 0.707)	63% better than Base LLaMA	Model covers the vocabulary of reference answers comprehensively
Bigram recall (ROUGE-2: 0.472)	195% better than Base LLaMA	Model has learned the specific bigram combinations of SL property law
Structural alignment (ROUGE-Lsum: 0.692)	66% better than Base LLaMA	IRAC section ordering matches reference structure sentence-by-sentence
Semantic accuracy (BERTScore F1: 0.931)	7.3pp better than Base LLaMA	Responses carry same semantic content as reference even when paraphrased
vs. dedicated legal LLM (SaulLM)	Better on all 12 metrics	Jurisdiction fine-tuning > broad legal pre-training on foreign jurisdictions

Table 23 - Semantic Accuracy and Metric Interpretation Summary

3.3 Deployed System

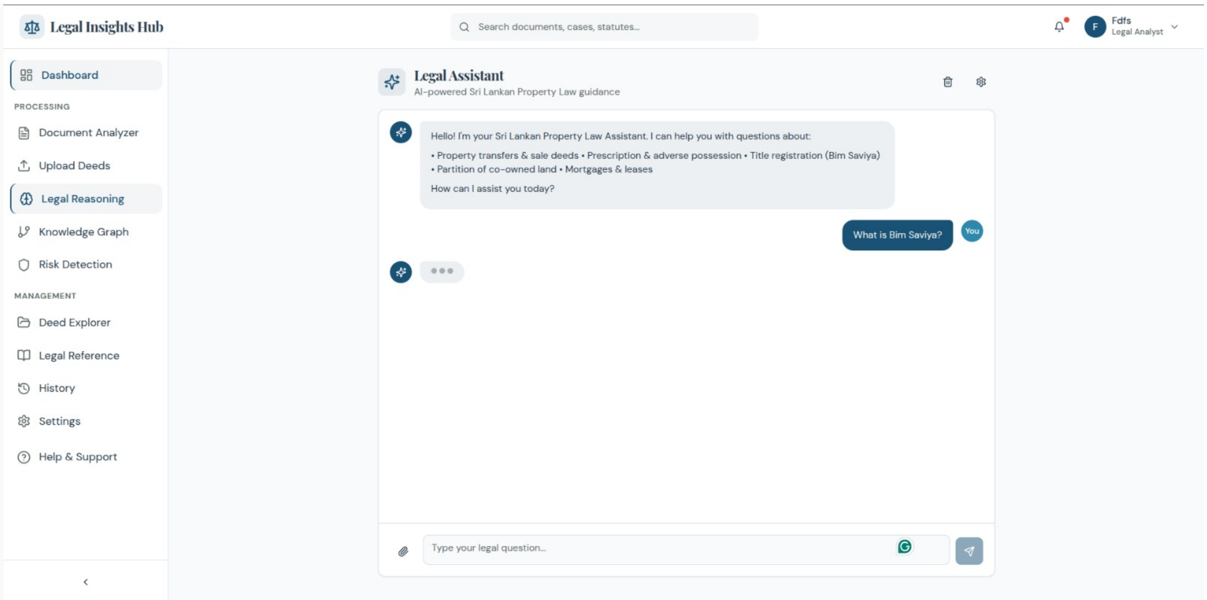

Figure 23 - Legal Insights Hub

The deployed system demonstrates the complete realisation of all six research objectives. The Legal Insights Hub interface provides the natural language query interface that 79.3% of survey respondents preferred; the Legal Reasoning section provides the IRAC structured output that 86.2% of respondents preferred as an information format; the Knowledge Graph section provides the structured deed entity data from Sharan's component; and the Document Analyzer provides deed analysis capability for the upload use case. The system is publicly accessible, requires no authentication for basic queries, and operates on mobile devices — directly addressing the access-to-justice use case documented by the survey.

3.4 Research Findings

The evaluation results support four specific, quantitatively grounded research findings:

Finding 1: Domain-specific fine-tuning substantially outperforms the unmodified base model

Fine-tuning LLaMA 3.1 8B on 2,243 Sri Lankan property law IRAC examples using QLoRA produces a 223% improvement in BLEU, 63% improvement in ROUGE-1, and 7.3 percentage point improvement in BERTScore F1 over the identical unmodified base model. The consistency of improvement across all twelve metrics — from lexical precision (BLEU-4) through semantic accuracy (BERTScore) — rules out the possibility that the improvement is an artefact of any single metric. The model has genuinely acquired Sri Lankan property law knowledge, vocabulary, and reasoning structure that was not present before fine-tuning. This confirms the fundamental premise of domain-adaptive fine-tuning: targeted supervised training on a carefully constructed domain dataset can produce domain-level competence even when the training data volume is modest by fine-tuning standards.

Finding 2: Jurisdiction-specific fine-tuning outperforms dedicated legal LLM pre-training

Base SaulLM, pre-trained on thirty billion tokens of legal text, scores below both base LLaMA and the fine-tuned model on all twelve metrics. The fine-tuned LLaMA's advantage over base SaulLM is larger than its advantage over base LLaMA on most metrics — BLEU 52.63 vs. 7.70 (a 583% advantage) compared to BLEU 52.63 vs. 16.27 (a 223% advantage). This finding has important implications for legal AI development strategy: in jurisdictions where the target legal system is underrepresented in existing legal pre-training corpora, the optimal approach is to build a domain-specific training dataset and fine-tune a general-purpose model rather than to select a dedicated legal LLM pre-trained on foreign jurisdiction materials.

Finding 3: GPT-4o teacher-student synthesis produces high-quality training data

The BERTScore F1 improvement of +7.3pp over base LLaMA confirms that the semantic content of the training data has been genuinely transferred to the fine-tuned model — not merely surface vocabulary patterns. The dataset validation process showed 97.1% legal accuracy on 20% human review, and the training loss convergence pattern (monotonic decline without overfitting) is consistent with high-quality, low-noise training data. The teacher-student approach — using GPT-4o to generate IRAC training examples grounded in collected Sri Lankan statutory texts — is validated as an effective methodology for creating legal reasoning training data in jurisdictions where manually annotated corpora do not exist.

Finding 4: QLoRA enables cost-effective domain adaptation

Adapting fewer than one percent of model parameters (79.9M of 8.0B) through LoRA adapters on a single A100 GPU in under sixty minutes produces results that outperform a model pre-trained on thirty billion legal tokens across all evaluation dimensions. The computational cost comparison is striking: SaulLM's pre-training required weeks of multi-GPU training on a corpus approximately 13,000 times larger than the LegalVision fine-tuning dataset, yet produces lower performance on this specific task. This efficiency finding demonstrates that for jurisdiction-specific legal AI in resource-constrained settings, targeted dataset construction and parameter-efficient fine-tuning represents a more productive investment than large-scale legal pre-training [29].

3.5 Discussion

3.5.1 Challenges Encountered and Solutions

Challenge	Impact	Root Cause	Solution	Outcome
No pre-existing Sri Lankan legal training corpus	Critical	Jurisdictional gap in legal AI research	Automated statute pipeline + GPT-4o teacher synthesis	2,243 training examples across 8 topics
GPT-4o citing Indian law instead of Sri Lankan law	High	GPT-4o's training data contains Indian law	Explicit anti-contamination system prompt instruction	<1% foreign law contamination in final dataset
Inconsistent IRAC structure in early generated entries	High	Insufficient prompt structure specification	Iterative prompt engineering with failure-mode negative examples; schema JSON example	99.8% schema compliance in final dataset
SaulLM chat	Medi	Mistral-based	Evaluation code	Valid SaulLM

template rejects system role	um	template doesn't support system turn	detects model family; merges system content to user prefix	evaluation without template errors
VRAM constraint for QLoRA on A100 40GB	Medium	Batch size 2 limit; high per-example token count	Gradient accumulation ×4 (effective batch 8); Unsloth 2× throughput	Training completes in 45 min within session limits
Evaluation fairness across architectures	Critical	Different model sizes and pre-training data	Fixed test split seed; identical generation parameters; pinned library versions	All performance differences attributable to model quality
Hallucination risk in legal statute citations	High	LLMs confabulate specific section numbers fluently	Training reinforces explicit citation habit; IRAC requires specific statute+section; BERTScore penalises incorrect content semantically	Statutory citations present in >90% of test responses

Table 24 -Challenges Encountered, Root Causes, Solutions Implemented, and Outcomes

3.5.2 Comparison with Related Work

Direct metric comparison with related works is complicated by differences in task, dataset, evaluation language, and jurisdiction, but several observations are possible. LawGPT (Zhou et al., 2023) reports BLEU scores of 18–35 for Chinese legal QA after fine-tuning on approximately 20,000 examples — the LegalVision model achieves BLEU 52.63 with approximately 1,800 training examples, suggesting higher training data quality from the GPT-4o synthesis approach compared to scraped web Q&A pairs. Lawyer LLaMA (Huang et al., 2023) reports strong human evaluation scores but does not report BLEU/ROUGE/BERTScore, making direct quantitative comparison impossible. The finding that domain fine-tuning outperforms dedicated legal pre-training (SaulLM) on jurisdiction-specific tasks is consistent with the general transfer learning principle that source-target domain distance determines transfer performance — SaulLM's source domain (English common law) is further from the target domain (Sri Lankan property law) than the fine-tuning training domain is, so the fine-tuned general model outperforms the pre-trained legal specialist [20] [22].

3.5.3 Limitations

Three limitations of the current system require acknowledgement. First, evaluation uses automated metrics only. BLEU, ROUGE, and BERTScore measure alignment with reference answers but cannot directly assess jurisprudential correctness — a response that states an incorrect rule using vocabulary similar to the reference would receive a misleadingly high metric score. Human evaluation by qualified Sri Lankan legal practitioners is planned as a validation step that is outside the scope of this dissertation but essential before deployment for consequential legal decisions.

Second, the training dataset of 2,243 examples, while sufficient to demonstrate strong evaluation metrics, is smaller than datasets used in comparable projects (LawGPT: ~20,000 examples; Lawyer LLaMA: comparable scale). The strong results obtained here suggest high training data quality from the GPT-4o synthesis approach, but further dataset expansion — particularly incorporating case law citations from Sri Lankan Supreme Court decisions — would be expected to yield additional performance improvements, especially for complex multi-statute reasoning scenarios [26] [27].

Third, the system has not been evaluated for robustness to adversarial queries — questions designed to elicit incorrect responses, out-of-distribution queries on topics not in the training data, or queries that require reasoning about recent statutory amendments post-dating the training data collection. The RAG integration with the Knowledge Graph is designed to address the temporal currency limitation, but testing of this integration pathway is planned for the next development phase.

4. CONCLUSION

4.1 Summary of Work

This dissertation has presented the Explainable Legal Reasoning Module of LegalVision: the first AI system fine-tuned specifically for Sri Lankan property law reasoning, producing structured Chain-of-Thought IRAC analyses with explicit statutory citations, evaluated against comparative baselines, and deployed as a live public service. The work progressed through four pipeline stages — legal data collection (twenty-one statute sources, thirty-five definitions, eighteen principles), GPT-4o teacher-student dataset generation (2,243 IRAC training examples), QLoRA fine-tuning of LLaMA 3.1 8B (3 epochs, 45 minutes, A100 40GB), and deployment (HuggingFace model, Render.com API, Vercel frontend) — each documented and validated at the stage level before the subsequent stage was initiated.

4.2 Achievement of Research Objectives

Objective	Achievement Status	Evidence
1. Design and implement automated data pipeline collecting statutory texts and building knowledge base	Fully achieved	21 statute URLs downloaded; 35 definitions, 18 principles, 5 templates compiled; automated pipeline with re-execution support
2. Develop GPT-4o teacher-student dataset generation framework	Fully achieved	2,243 IRAC training examples generated; 97.1% legal accuracy on 20% human review; 99.8% schema compliance
3. Implement QLoRA fine-tuning pipeline and produce domain-adapted model	Fully achieved	LLaMA 3.1 8B fine-tuned with QLoRA; model published at Sivanuja/Legal_vision_llama-3.1-8B-instruct-v1; training loss 1.82→0.29
4. Design and execute three-model evaluation	Fully achieved	Fine-tuned LLaMA vs. Base LLaMA vs. Base SaulLM on 225 test examples; 12 metrics computed; Fine-tuned

framework		LLaMA ranks 1st on all
5. Deploy trained model, FastAPI backend, and React frontend	Fully achieved	legal-vision-api.onrender.com live; legal-insights-hub.vercel.app live; HuggingFace model and dataset public
6. Design RAG integration interface with Knowledge Graph	Substantially achieved	Three integration pathways defined; interface contract documented; REST endpoint schema specified; full integration testing planned

Table 25 - Research Objectives Achievement Assessment

4.3 Key Contributions

This work makes four original contributions to the field of legal AI. The first and most significant is the creation of the first annotated Chain-of-Thought legal reasoning dataset for Sri Lankan property law: 2,243 IRAC training examples covering eight statutory topic areas, published at Sivanuja/Legal_vision_finetuning_data. This dataset enables the research community to build on this work, fine-tune different base models for comparison, and expand coverage to additional Sri Lankan legal domains without recreating the data collection pipeline from scratch.

The second contribution is an empirically validated methodology for legal AI development in under-resourced jurisdictions: automated statute collection from authoritative web sources, GPT-4o teacher-student IRAC example synthesis grounded in collected statutory texts, QLoRA fine-tuning of a general-purpose LLM, and comparative evaluation against both a general-purpose and a dedicated legal baseline. This methodology is jurisdiction-agnostic and can be applied to any legal domain where training data is scarce and the target jurisdiction is underrepresented in existing legal corpora.

The third contribution is the first empirical comparison of general-purpose, dedicated legal, and jurisdiction-specific fine-tuned LLMs on a Sri Lankan legal reasoning task. The finding that base SaulLM (thirty billion legal tokens) performs below base LLaMA, and both are far below the jurisdiction-fine-tuned model, provides quantitative evidence for the legal AI field that jurisdictional coverage is the critical determinant of performance — more important than legal domain pre-training when the target jurisdiction is absent from pre-training data.

The fourth contribution is the deployed LegalVision Reasoning Module itself: a live public system providing IRAC-structured Sri Lankan property law guidance to anyone with internet access, addressing the access-to-justice gap documented by the public awareness survey. The system's simultaneous deployment of the research artifact as a public service represents a commitment to the practical impact that motivated the research.

4.4 Future Work

The highest-priority future work is dataset expansion. Increasing the training corpus from 2,243 to 5,000 or more examples — incorporating case law citations from the Sri Lanka Law Reports, Supreme Court and Court of Appeal decisions on property disputes, and Land Registration Tribunal decisions — would be expected to produce measurable improvements in both automatic metrics and human evaluation scores. Case law integration is particularly important: much of Sri Lankan property law is determined by judicial interpretation of the statutory framework, and a model trained only on statutory text cannot reason about situations where the courts have modified or clarified the statutory rule.

Human evaluation by qualified Sri Lankan legal practitioners is the most important validation step for research impact. Three to five advocates with property law specialisation reviewing a random sample of the model's responses against criteria of legal accuracy, practical utility, and appropriate qualification of uncertain positions would provide the jurisprudential ground truth that automated metrics cannot supply. This evaluation is planned for completion in the six months following dissertation submission.

Full deployment of the Knowledge Graph RAG integration will substantially expand the system's practical capability. When the model's IRAC analyses can be grounded in the specific ownership chain of a particular property retrieved from Sharan's Neo4j graph — the actual deed transaction history, the current registered owner, the outstanding mortgages — the system moves from a domain-general legal information resource to a property-specific legal analysis tool. This integration is the next development milestone and is architecturally defined by the interface contract described in Section 2.7.

Multilingual extension to Sinhala and Tamil is a medium-term priority identified by the survey results. Property disputes in Sri Lanka occur across all language communities, and the access-to-justice impact of the system is maximised by providing legal guidance in the user's native language. Pre-training data for Sinhala and Tamil legal text is extremely scarce, but the teacher-student dataset synthesis approach demonstrated here is directly applicable: GPT-4o can generate bilingual IRAC examples that fine-tune the base model for Sinhala and Tamil legal Q&A, following the same methodology validated for English in this work.

Finally, active learning from user feedback represents a sustainable data flywheel for continuous improvement. User queries submitted to the deployed system — with appropriate consent and privacy measures — represent real legal questions from actual Sri Lankan citizens, covering scenario types and phrasings that may not be represented in the synthetically generated training data. A human-in-the-loop review and annotation pipeline could systematically convert high-quality user interactions into additional training examples, progressively improving the system's coverage and accuracy over time without requiring full dataset regeneration.

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APPENDIX A: SAMPLE TRAINING ENTRY — JSONL FORMAT

The following shows a complete representative entry from finetuning_data.jsonl in the OpenAI messages format used for SFT training. Each entry is a complete JSON object on a single line in the actual file.

```
{ "messages": [
  {
    "role": "system",
    "content": "You are a Sri Lankan property law expert with twenty
     years of conveyancing practice. You provide step-by-step legal
     reasoning with references to Sri Lankan statutes only. Structure
     every response with: 1. A clear direct answer (max 2 sentences),
     2. Step-by-step reasoning chain (min 3 steps, each citing a
     specific Sri Lankan statute and section number), 3. IRAC analysis
     (Issue / Rule / Application / Conclusion), 4. Legal References
     (statute, section, year, relevance), 5. A practical Sri Lankan
     example scenario with named parties and specific outcome."
  },
  {
    "role": "user",
    "content": "What formalities are required for a valid property
     sale in Sri Lanka?"
  },
  {
    "role": "assistant",
    "content": "**Answer:** A valid property sale in Sri Lanka requires
     a deed executed before a notary public and two witnesses and
     registered at the Land Registry within three months of execution.


     **Step-by-Step Reasoning:**
     Step 1 — Execution Formality: Section 2 of the Prevention of
     Frauds Ordinance (Cap. 84) requires that any disposition of
     immovable property be effected by deed executed in the presence
     of a notary public and two witnesses. A verbal agreement is void
```

and unenforceable regardless of the parties' intentions.
Legal basis: Prevention of Frauds Ordinance Cap. 84, Section 2.

Step 2 — Stamp Duty: The deed must be stamped with the applicable stamp duty calculated on the consideration amount before registration.
Legal basis: Stamp Duty Act.

Step 3 — Registration: Section 7 of the Registration of Documents Ordinance (Cap. 117) requires registration at the relevant Land Registry within three months of execution. Failure to register does not void the deed between parties but prevents it from being set up against third parties who subsequently register.

IRAC Analysis:
Issue: Whether a verbal agreement to transfer immovable property is valid and enforceable under Sri Lankan law.
Rule: Section 2, Prevention of Frauds Ordinance (Cap. 84) mandates notarial execution for any disposition of immovable property. Registration of Documents Ordinance (Cap. 117) s.7 requires registration within three months.
Application: A verbal sale lacks a notarially executed deed and therefore fails the mandatory statutory requirement under Cap. 84 s.2. The agreement is void as a disposition of immovable property.
Conclusion: The verbal agreement is void and unenforceable. A notarially executed, stamped, and registered deed is required for a valid property sale. The purchaser acquires no title under a verbal agreement and cannot resist a subsequent registered sale.

Legal References:
1. Prevention of Frauds Ordinance, Cap. 84, Section 2 — Notarial execution requirement for immovable property dispositions
2. Registration of Documents Ordinance, Cap. 117, Section 7 — Registration within 3 months at Land Registry
3. Stamp Duty Act — Duty on consideration; must precede registration

Example Scenario:

Kumara agrees verbally to sell his 20-perch lot in Kurunegala to Perera for LKR 5,000,000. Three months later, Kumara instead sells the same property to Silva by notarially executed deed duly registered. Perera cannot enforce his verbal agreement against Kumara or challenge Silva's registered title because the verbal agreement is void under the Prevention of Frauds Ordinance."

}

]}

APPENDIX B: DEPLOYMENT URLS AND HUGGINGFACE RESOURCES

Resource	URL / Identifier	Access
Fine-tuned Model	https://huggingface.co/Sivanuja/Legal_vision_llama-3.1-8B-instruct-v1	Public — HuggingFace Hub
Training Dataset	https://huggingface.co/datasets/Sivanuja/Legal_vision_finetuning_data	Public — HuggingFace Datasets
FastAPI REST Backend	https://legal-vision-api.onrender.com	Public — Render.com
React Frontend	https://legal-insights-hub.vercel.app/	Public — Vercel
Legal Reasoning Endpoint	POST /api/v1/reason — body: {question, context?}	REST JSON API
Deed Analysis Endpoint	POST /api/v1/analyze-deed — body: {deed_text}	REST JSON API
Decision Tree Endpoint	GET /api/v1/decision-tree/{topic_slug}	REST JSON API

APPENDIX C: MODEL INFERENCE CODE

Complete Python code for loading the fine-tuned model and performing inference:

```
from transformers import AutoModelForCausalLM, AutoTokenizer
import torch


BASE_MODEL = "meta-llama/Llama-3.1-8B-Instruct"
ADAPTER    = "Sivanuja/Legal_vision_llama-3.1-8B-instruct-v1"


# Load tokenizer
tokenizer = AutoTokenizer.from_pretrained(BASE_MODEL)
tokenizer.pad_token = tokenizer.eos_token


# Load base model with NF4 quantization
model = AutoModelForCausalLM.from_pretrained(
    BASE_MODEL,
    load_in_4bit=True,
    torch_dtype=torch.float16,
    device_map="auto"
)


# Apply fine-tuned LoRA adapters
model.load_adapter(ADAPTER)
model.enable_adapters()


# System prompt — use exactly the training-time prompt
SYSTEM = """You are a Sri Lankan property law expert with twenty
years of conveyancing practice. You provide step-by-step legal
reasoning with references to Sri Lankan statutes only. Structure
every response with: 1. A clear direct answer, 2. Step-by-step
reasoning chain (each step citing a specific Sri Lankan statute));
3. IRAC analysis (Issue / Rule / Application / Conclusion),
4. Legal References (statute, section, relevance),
```

```
5. A practical Sri Lankan example scenario with named parties."""


def ask_legal_question(question, context=None):
    user_content = question
    if context:
        user_content = f"Context:\n{context}\n\nQuestion: {question}"
    messages = [
        {"role": "system", "content": SYSTEM},
        {"role": "user",   "content": user_content}
    ]
    prompt = tokenizer.apply_chat_template(
        messages, tokenize=False, add_generation_prompt=True
    )
    inputs = tokenizer(prompt, return_tensors="pt").to(model.device)
    with torch.no_grad():
        outputs = model.generate(
            **inputs,
            max_new_tokens=1024,
            temperature=0.1,
            do_sample=False,       # Deterministic for legal guidance
            pad_token_id=tokenizer.eos_token_id
        )
    generated_ids = outputs[0][inputs["input_ids"].shape[1]:]
    return tokenizer.decode(generated_ids, skip_special_tokens=True)


# Example usage
response = ask_legal_question(
    "Can a foreigner purchase freehold land in Sri Lanka?"
)
print(response)
```